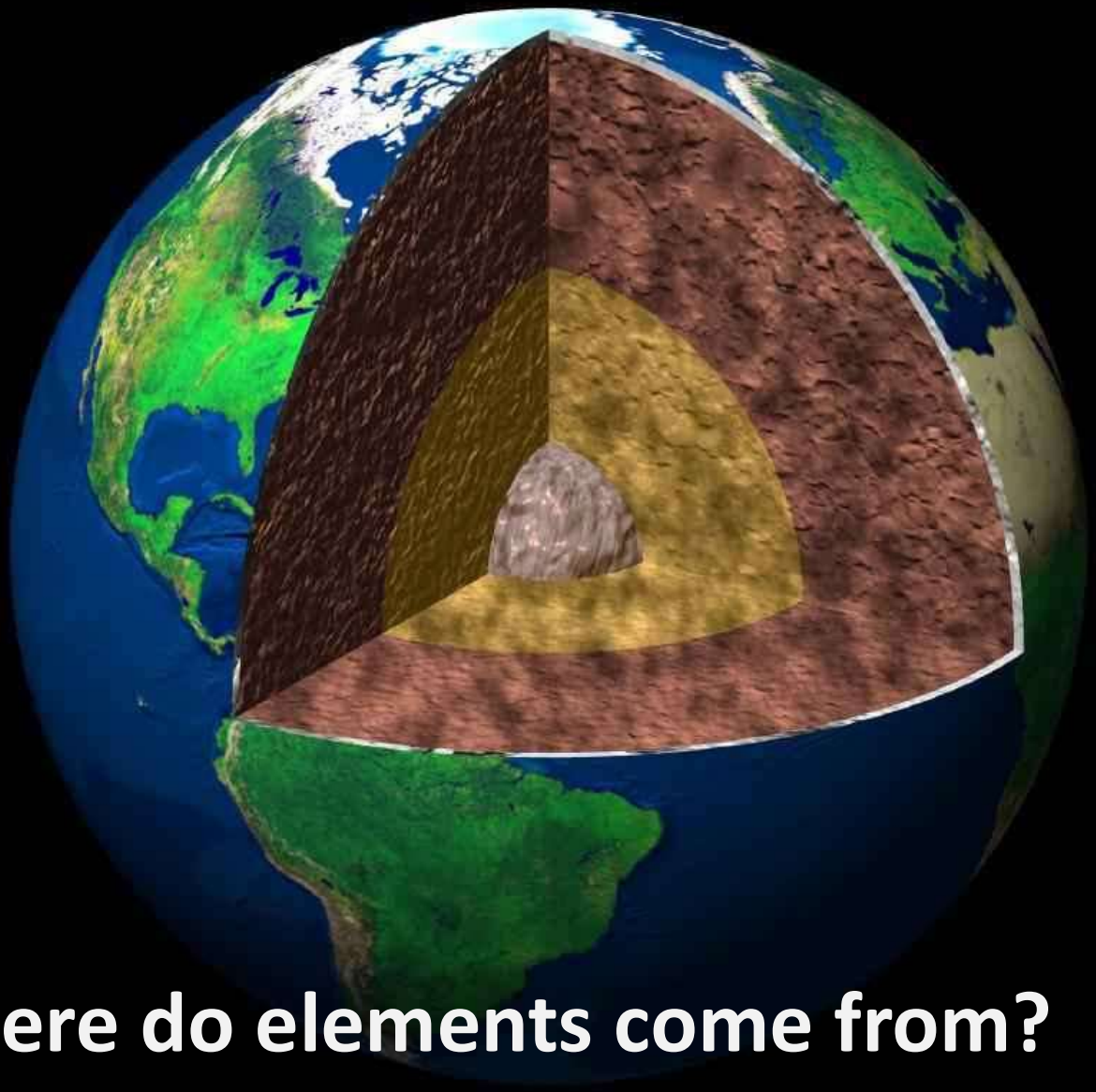
Chemistry of the Earth: Lecture 1

What is the Earth made of?

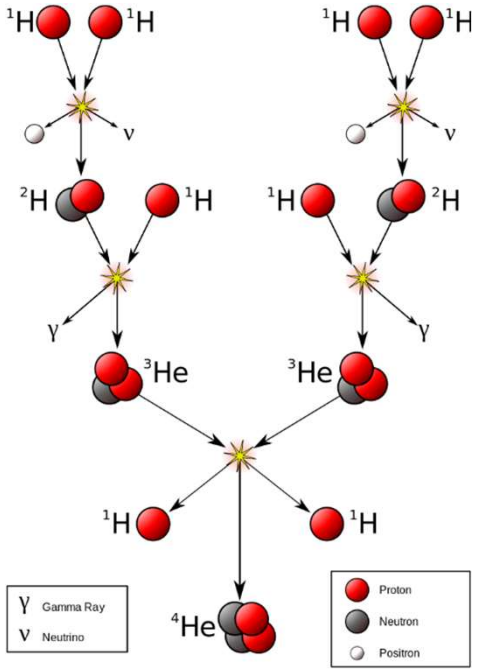


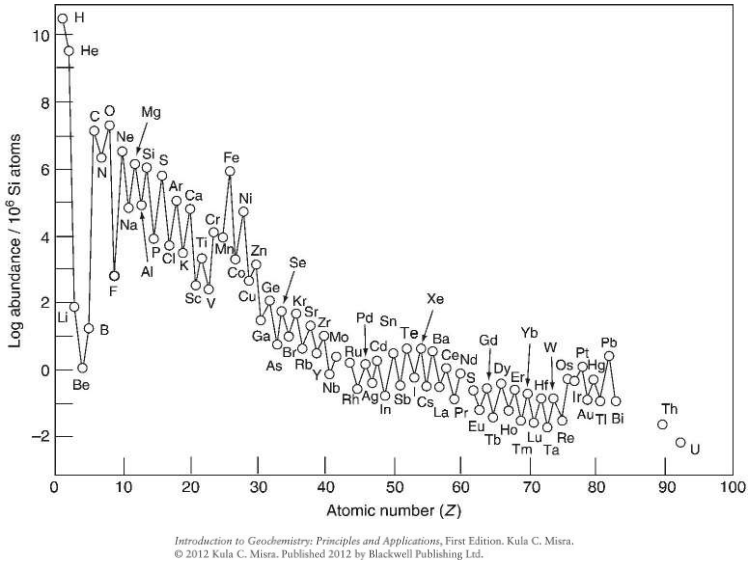
Part 1: Where do elements come from?

OBJECTIVES OF THIS WEEK'S LECTURE

- Review the basic structure of atoms, elements, and isotopes
- Understand how chemical elements are produced in stars
- Consider how the chemistry of the Earth relates to its place in the Universe and the Solar System







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THIS WEEK'S READING

Basic:

Grotzinger & Jordan **(Basic: Minimum reading)**	**Earth's layers: 7-12** **Structure of atoms: 55-56** **Formation of the Solar System: 600-602**
Marshak	Chapter 1

More detailed:

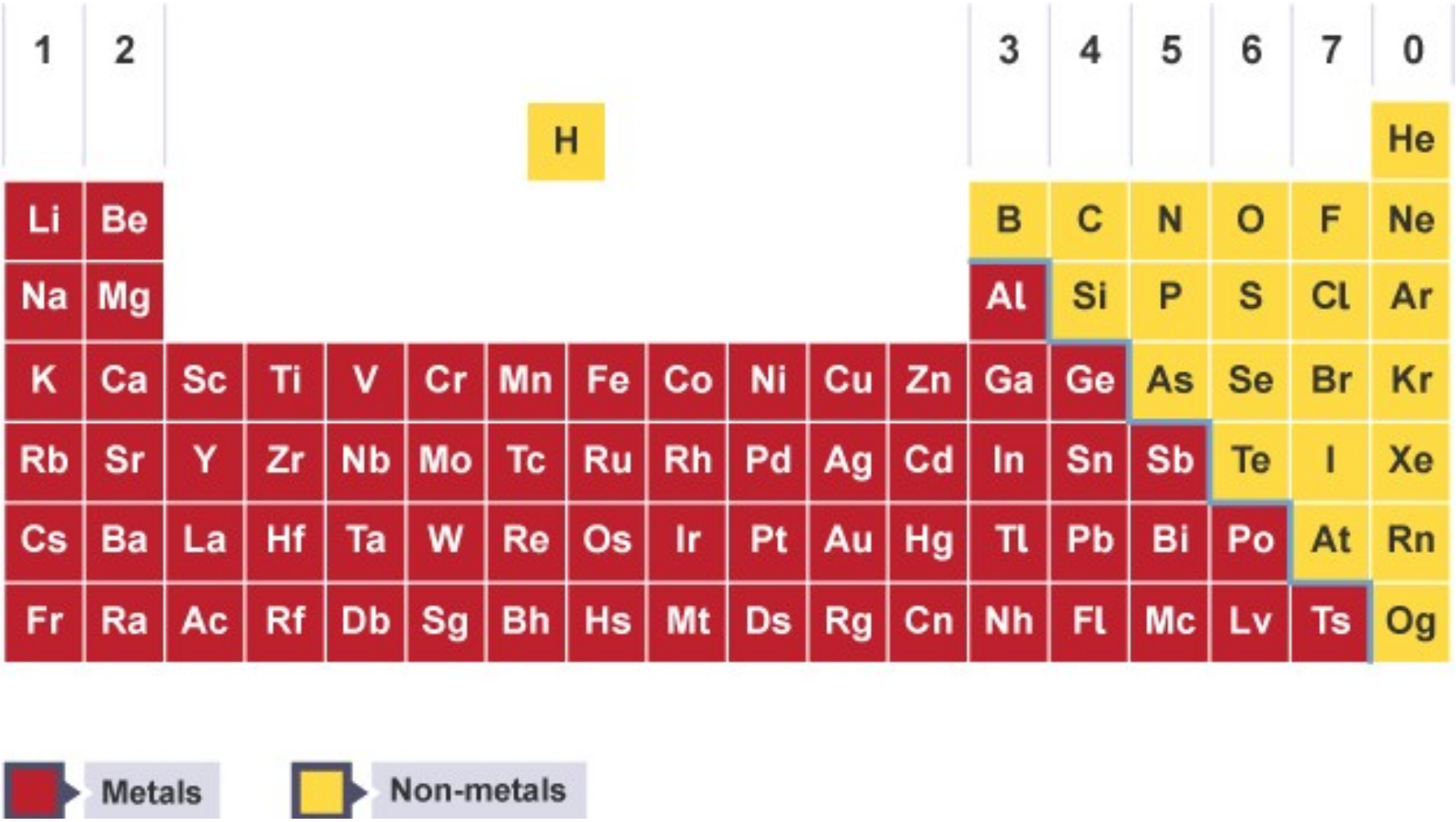
Gill	Chapter 11

More advanced:

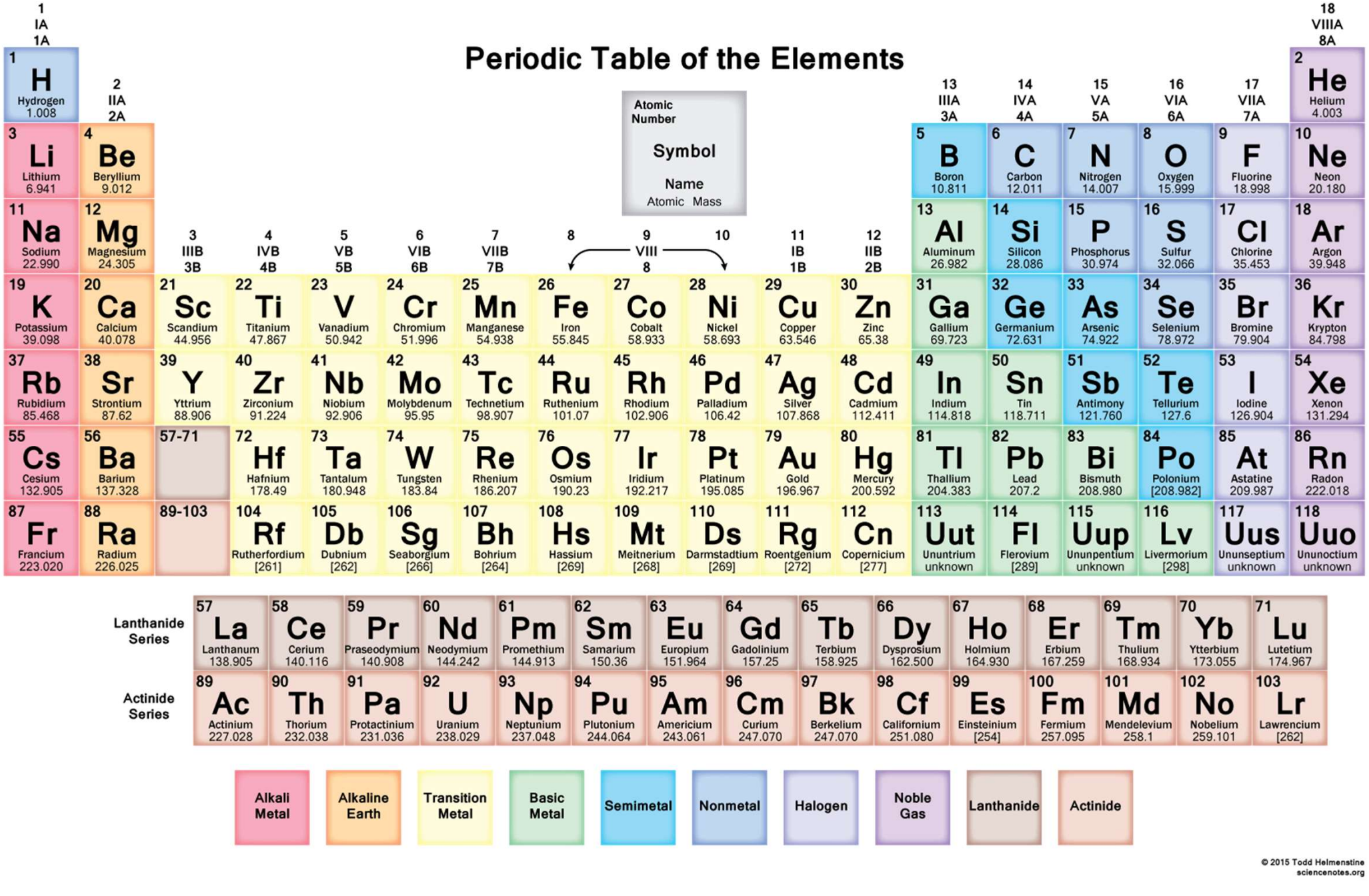
Misra	Chapter 12.1, Cosmic Perspective Chapter 12.2.2, Bulk Earth Composition
McSween & Richardson	Chapters 12 and 15
White	Chapters 1, 10, 11

THE PERIODIC TABLE OF THE ELEMENTS

GCSE Version:



http://www.bbc.co.uk/schools/gcsebitesize/science/aqa/fundamentals/atomsrev1.shtml



http://sciencenotes.org/printable-periodic-table/

Atomic Number

Symbol

Name

Atomic Mass

1

H

Hydrogen

1.008

3

Li

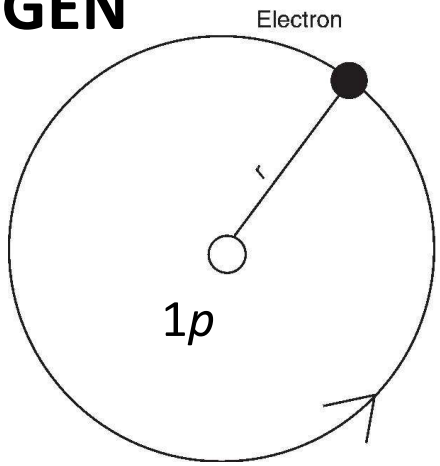
Lithium

6.941

HYDROGEN

^{1}H

Isotope mass (A)

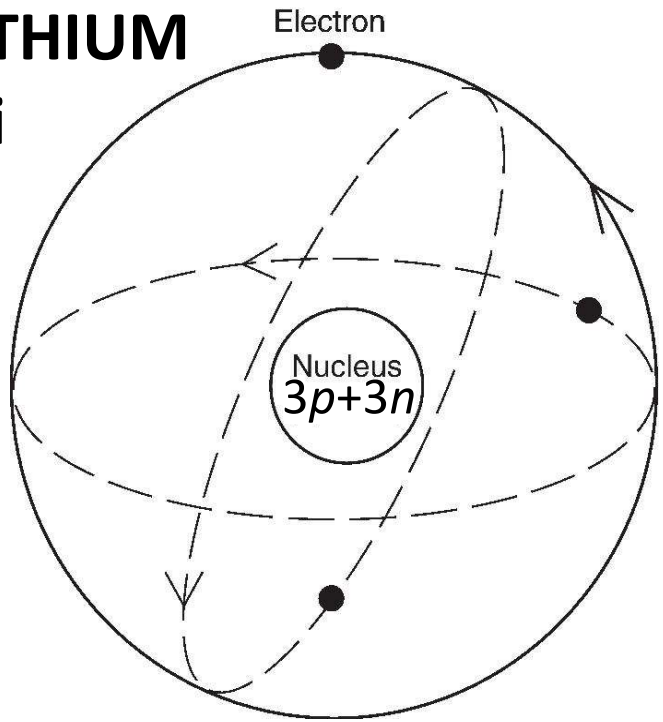


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LITHIUM

^{6}Li

Isotope mass (A)



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Atomic Number (Z)
Number of protons in nucleus.
Fixed for each element – number of protons defines the element.

Number of neutrons (N)
Number of neutrons in nucleus.
This can vary – **isotopes**.

Atomic Mass (A)
Number of protons + neutrons = Z+N = mass of nucleus.

Number of electrons
= number of protons *in a neutral atom.*
Variable numbers of electrons = chemistry!

Atomic Number

Symbol

Name

Atomic Mass

1

H

Hydrogen

1.008

3

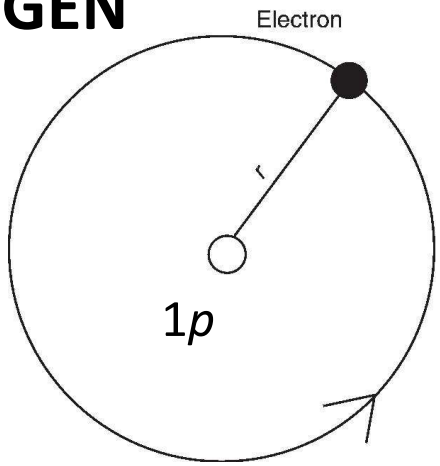
Li

Lithium

6.941

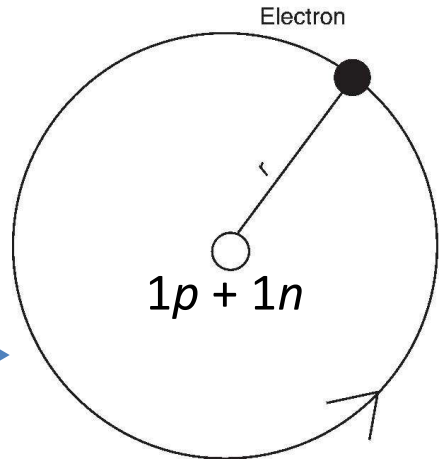
HYDROGEN

^{1}H



isotopes

^{2}H

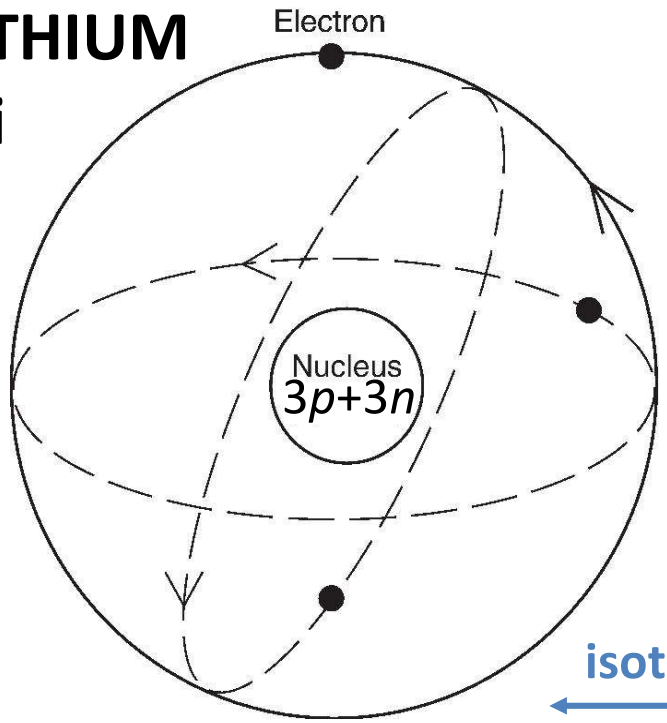


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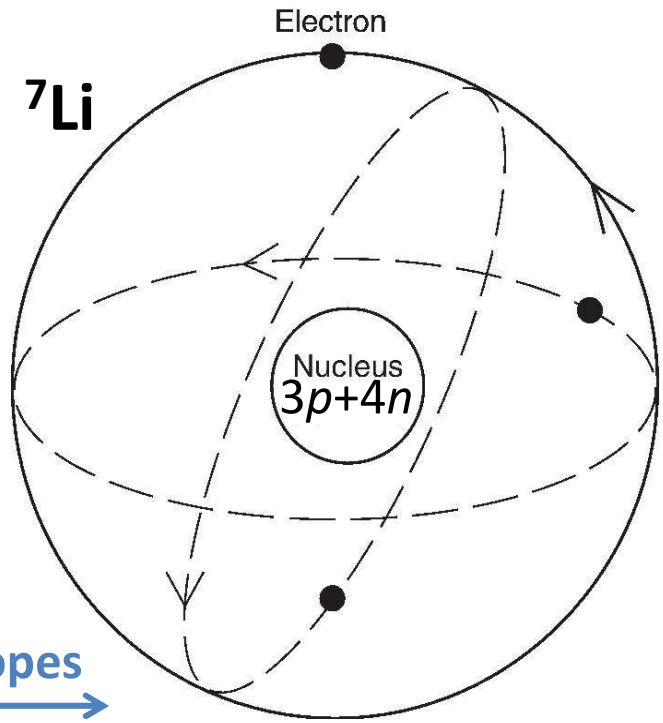
LITHIUM

^{6}Li



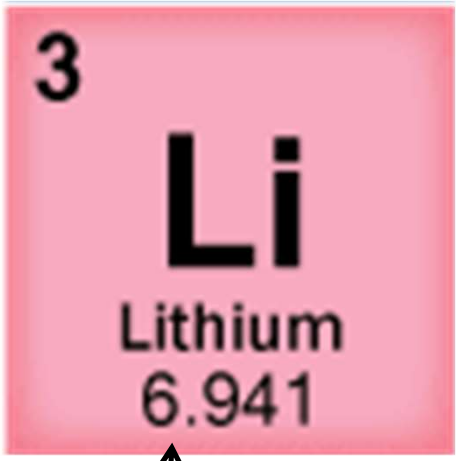
isotopes

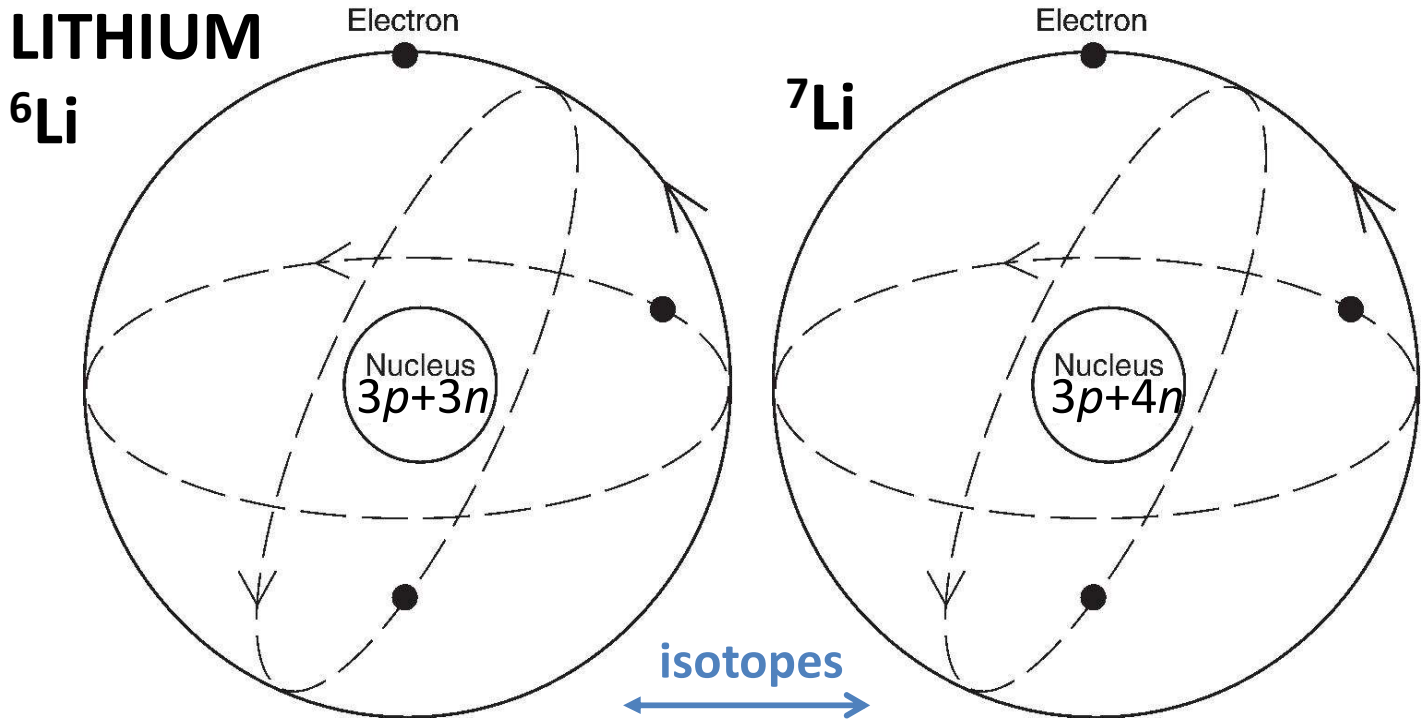
^{7}Li



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Atomic mass of element = average mass that occurs in nature.

Combines masses and abundances of isotopes:

^{7}Li = 92.4%

^{6}Li = 7.6%



http://periodictable.com/Elements/003/

WHERE DID THESE ELEMENTS COME FROM?
- PRODUCTION OF ELEMENTS IN STARS

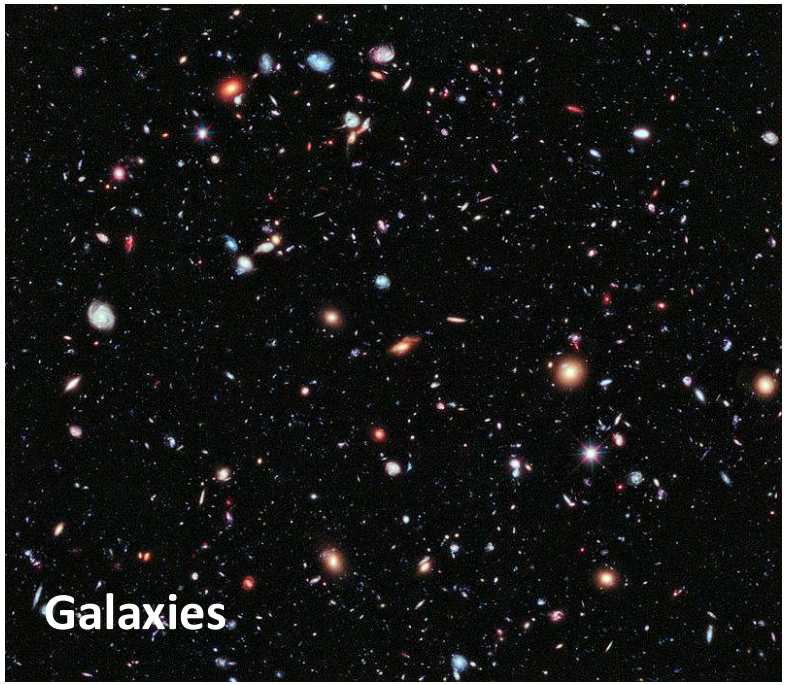
Formation of the Universe
Time = 0
Big Bang

Formation of first stars and galaxies

Formation of the Sun and Solar System
(SS Time = 0)

Evolution of stars and galaxies

13.8 Ga **4.6 Ga** **today**



Hubble Deep Field (HDF)
http://www.jpl.nasa.gov/spaceimages/details.php?id=PIA12110

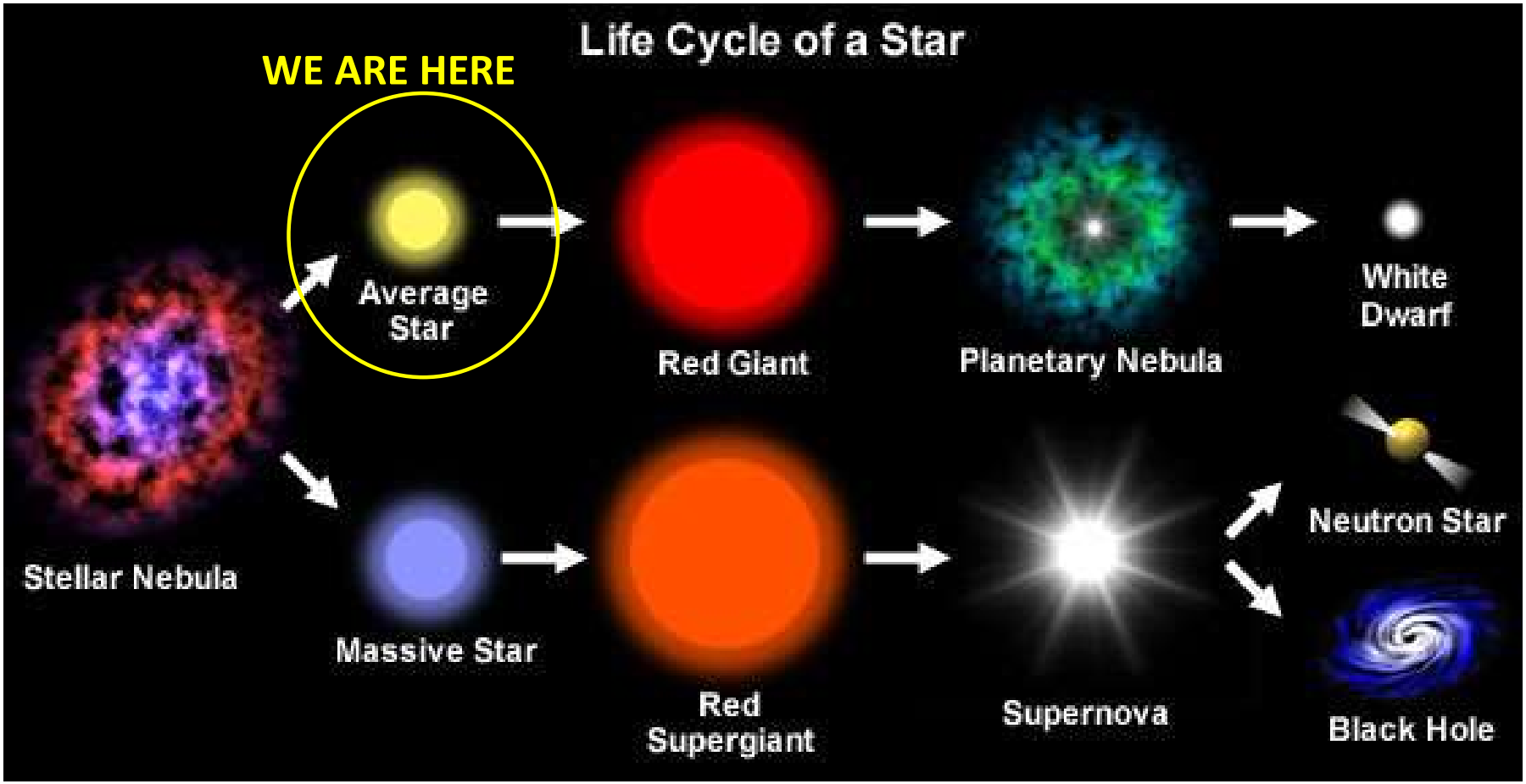


NASA

13.8 Ga = 13.8 billion years ago

LIFECYCLE OF STARS

- Big Bang produced hydrogen (H) and helium (He). These are the main ingredients.
- Collapse of a dense region of gas forms a new star.



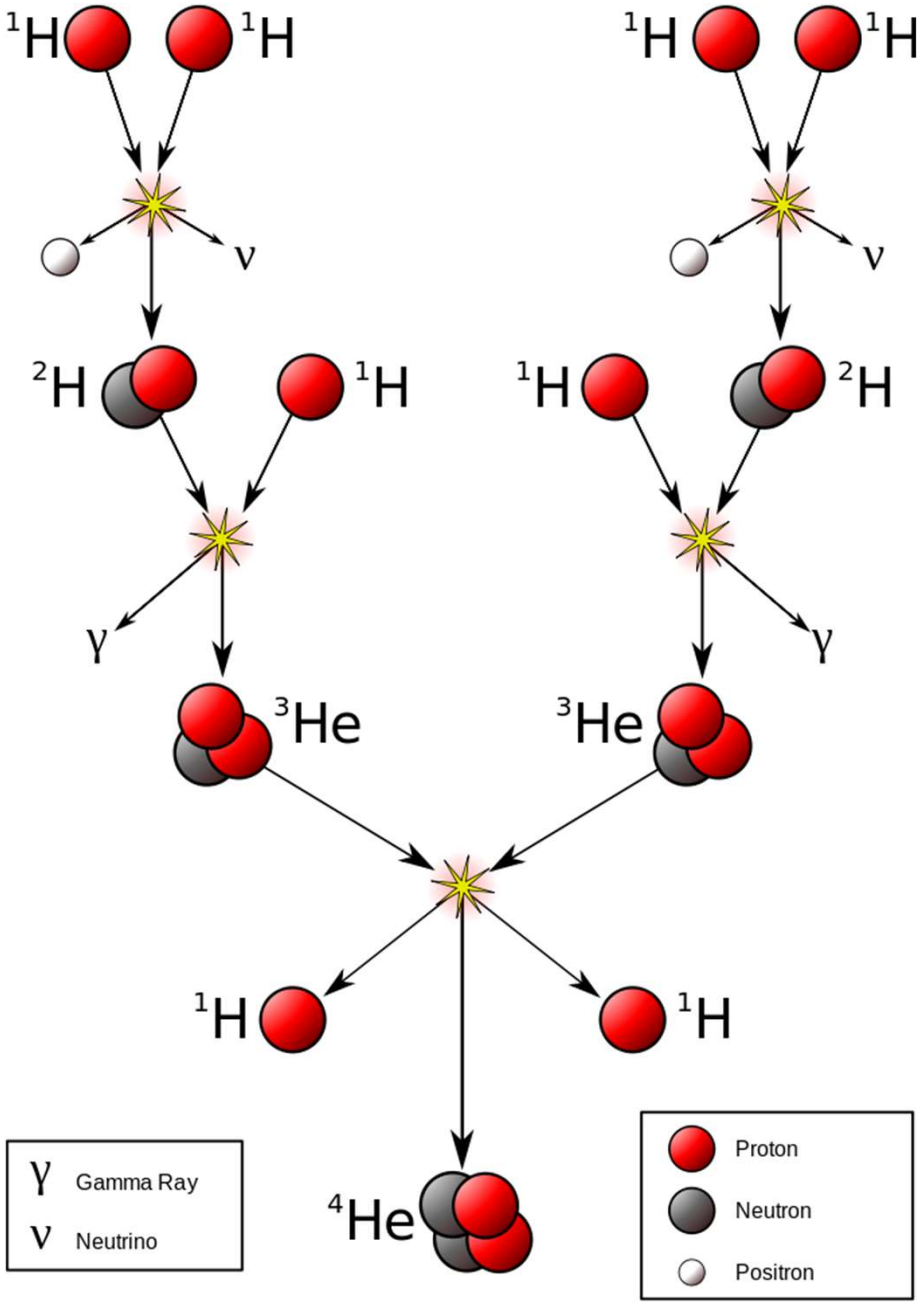
http://www.schoolsobservatory.org.uk/astro/stars/lifecycle

- When a star begins to “burn”, nuclear reactions begin (nucleosynthesis)
- Hydrogen burning - hydrogen atoms combine to make helium. Main phase of star’s lifetime. The Sun is about half way through this phase at the moment.

NUCLEOSYNTHESIS: CREATING NEW ELEMENTS

Hydrogen burning or proton-proton chain

- We only need to consider the nuclei of atoms.
- Hydrogen atoms (^1H) = protons.
- When two hydrogen atoms combine, one proton breaks down into a neutron and a positron.
- ^3He is produced first, then two ^3He nuclei combine to make one ^4He nucleus.

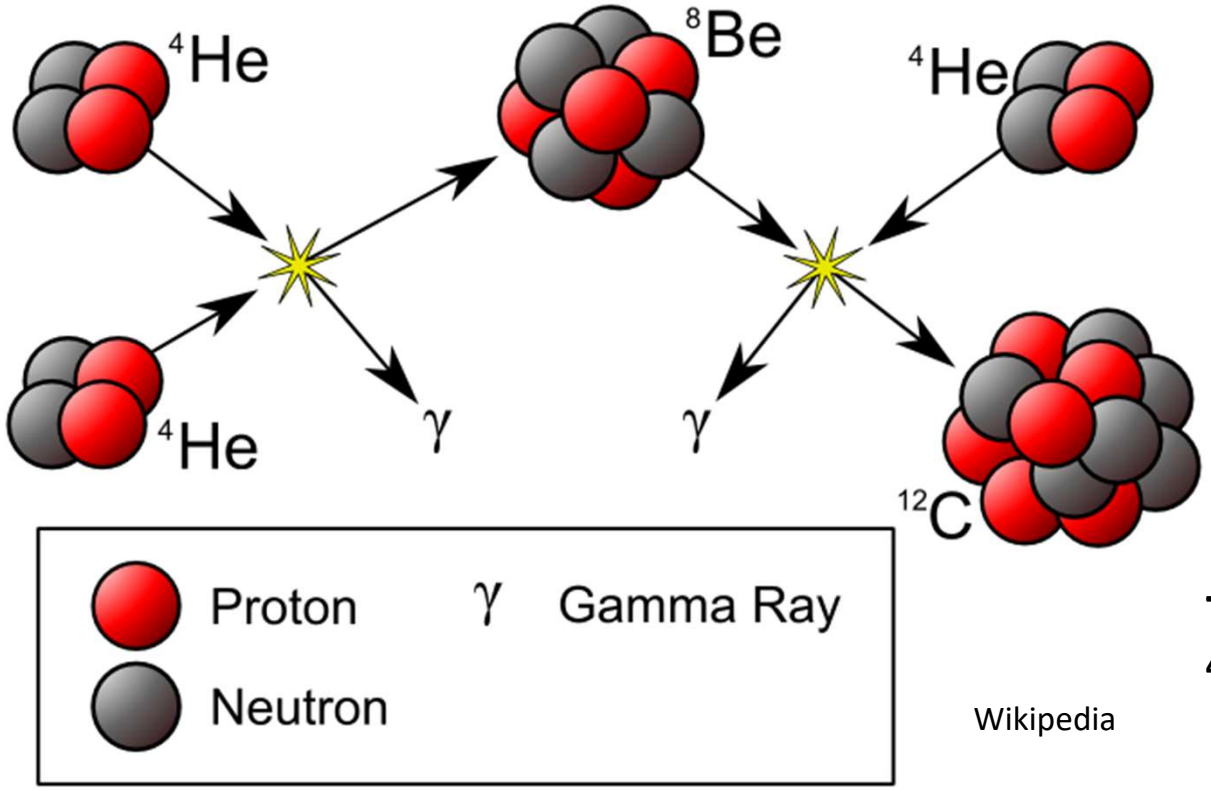


Wikipedia

Helium burning

• As a star evolves, helium builds up in the core. When hydrogen runs out, the core contracts and heats up. This makes the envelope expand – Red Giant.

• Helium nuclei begin to fuse together to produce ^{12}C.



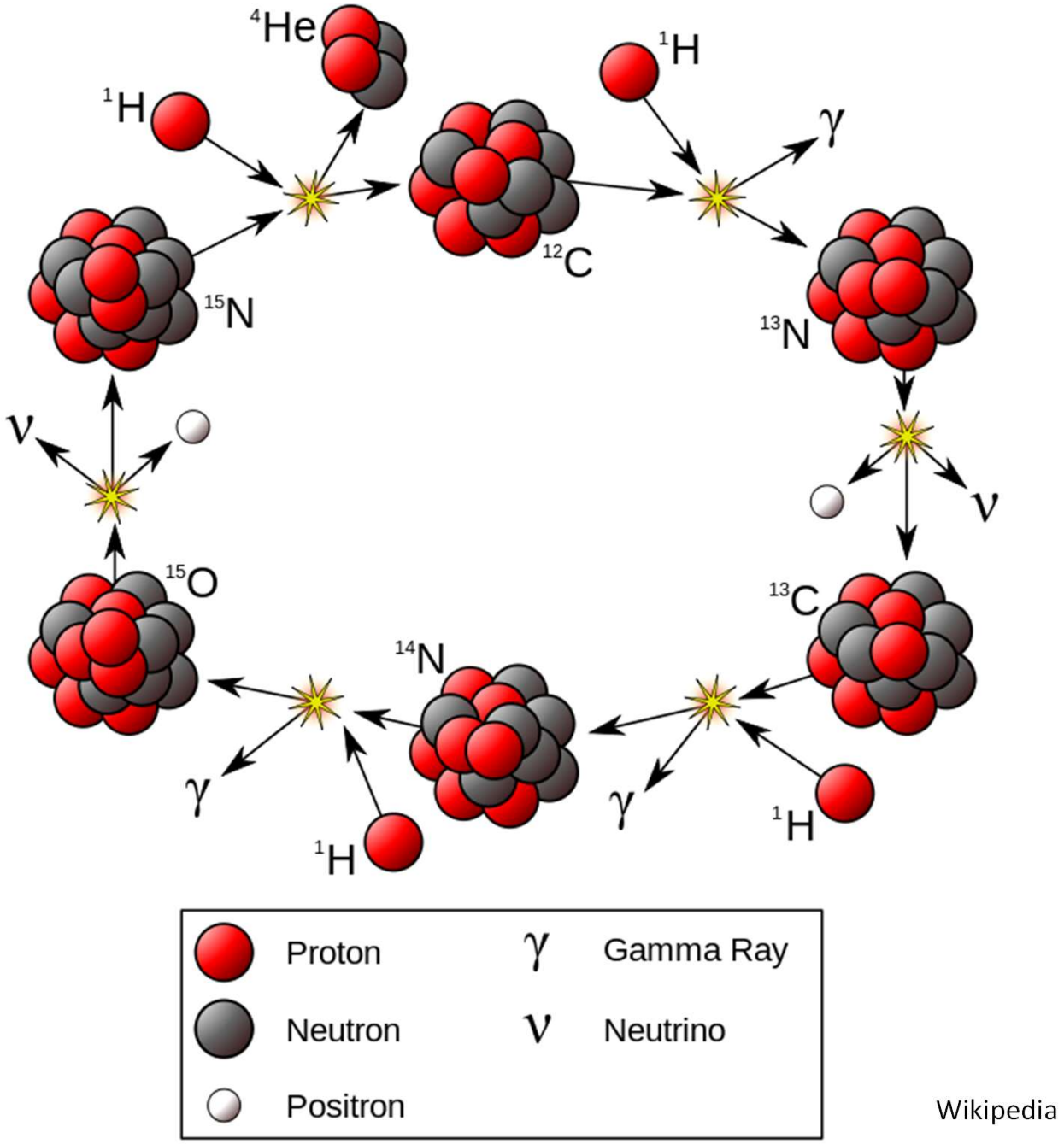
Wikipedia

Triple-alpha process
^{4}He = α particle

• Also, $^{12}C + {}^4He \rightarrow {}^{16}O$

The C-N-O cycle

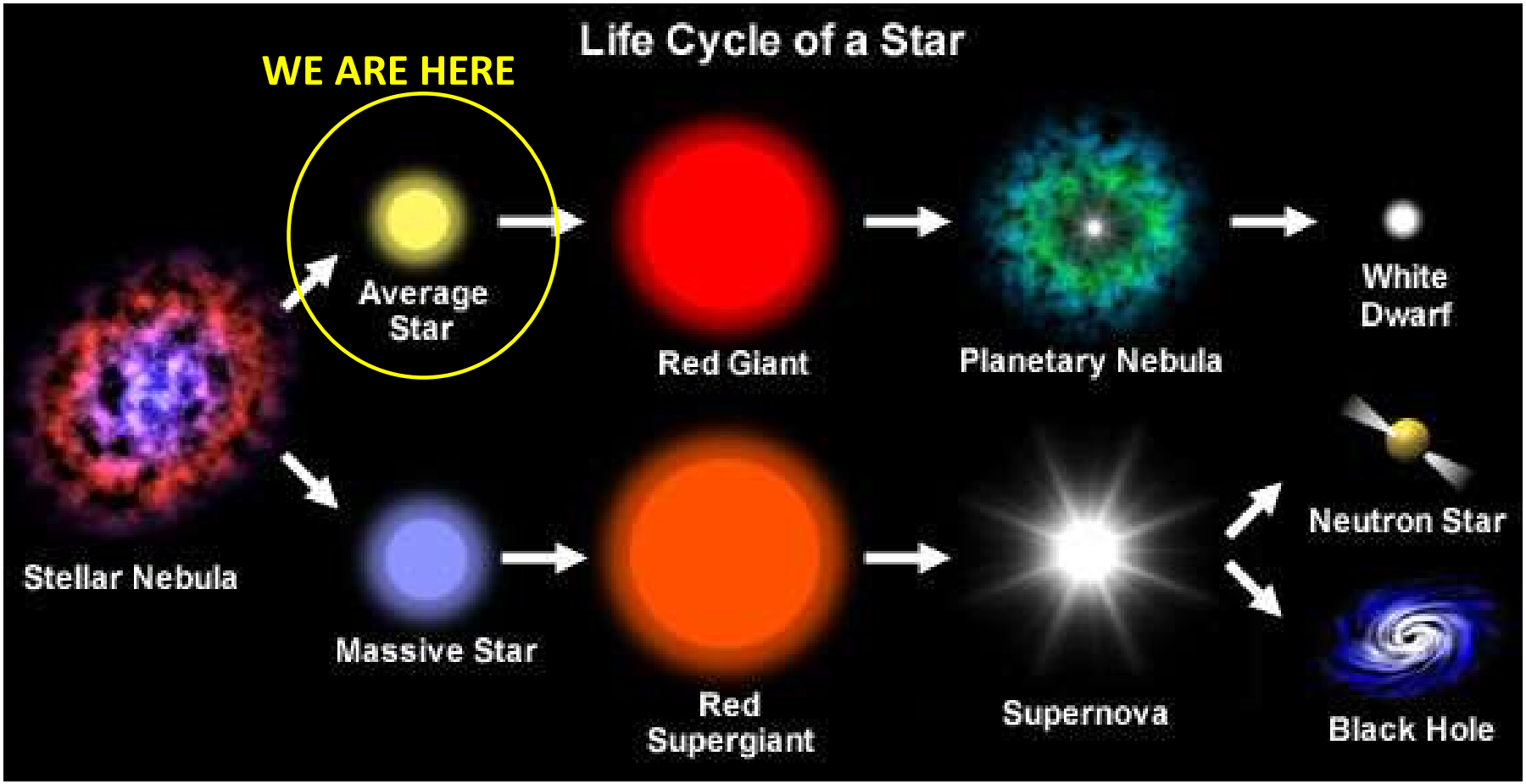
- Occurs in “second generation” stars that already contain ^{12}C produced in older stars
- Successive nuclear reactions produce He from 4 hydrogen nuclei, as well as ^{14}N (other nuclei are not stable)



Wikipedia

LIFECYCLE OF STARS

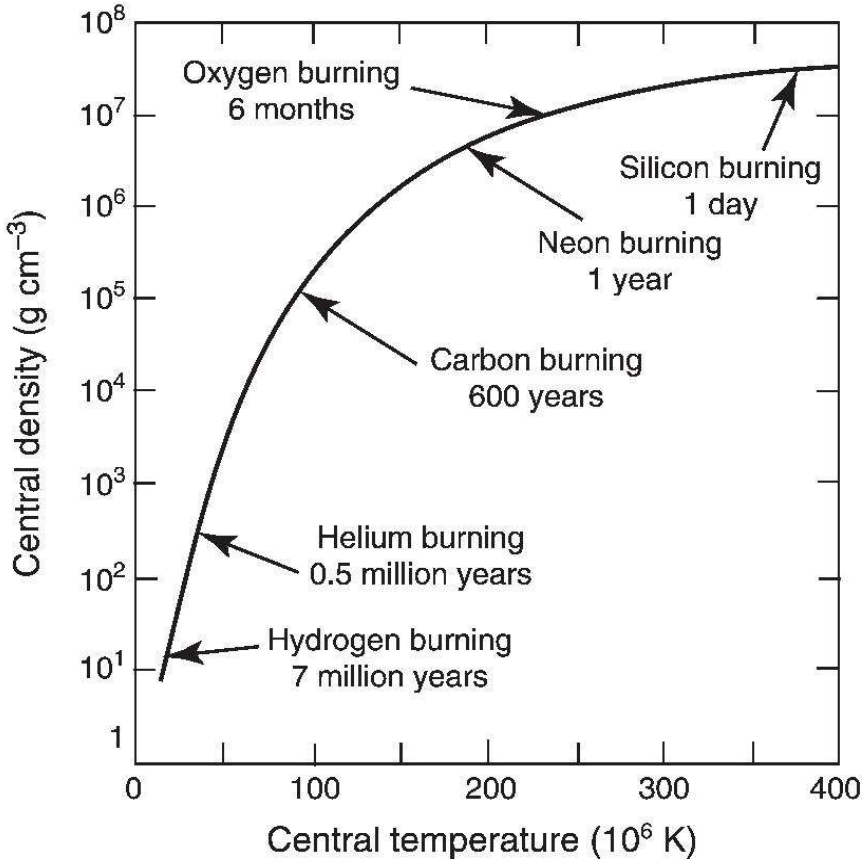
- Life and death of a star depends on its mass



http://www.schoolsobservatory.org.uk/astro/stars/lifecycle

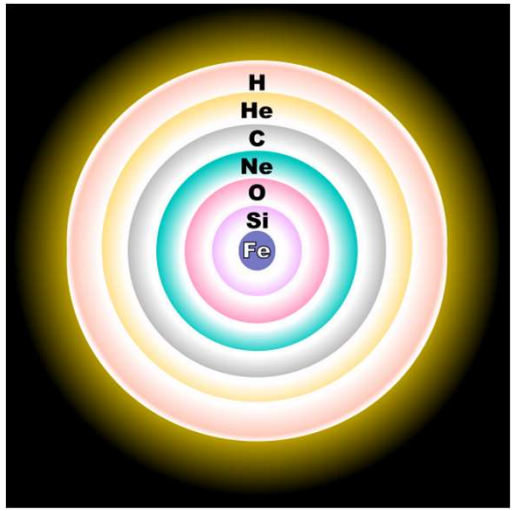
- Stars like the Sun end in a gently flowing wind of matter (planetary nebula) and eventually a white dwarf when burning ends. Lifetime ~10 billion years.
- Massive stars (~25 x Sun) end in explosive events like supernovae. Lifetime a few million years.

SUPERNOVAE



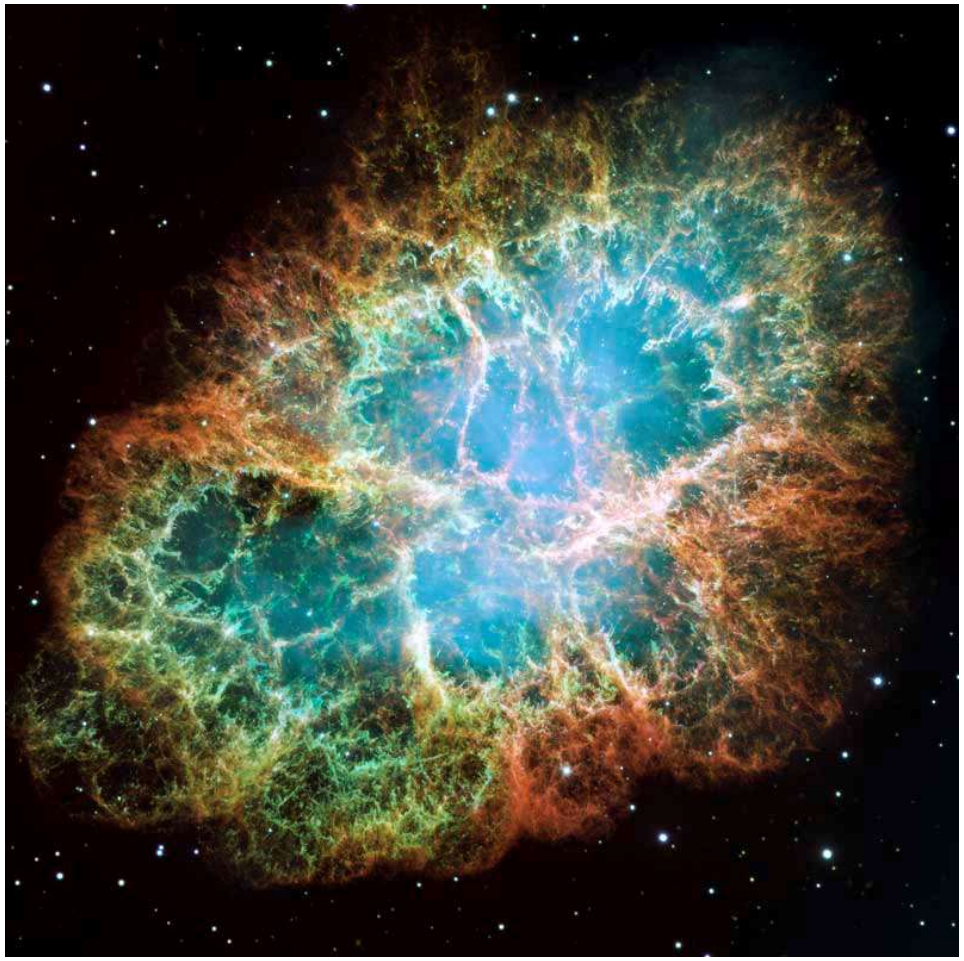
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Structure of a SN just before explosion



Wikipedia

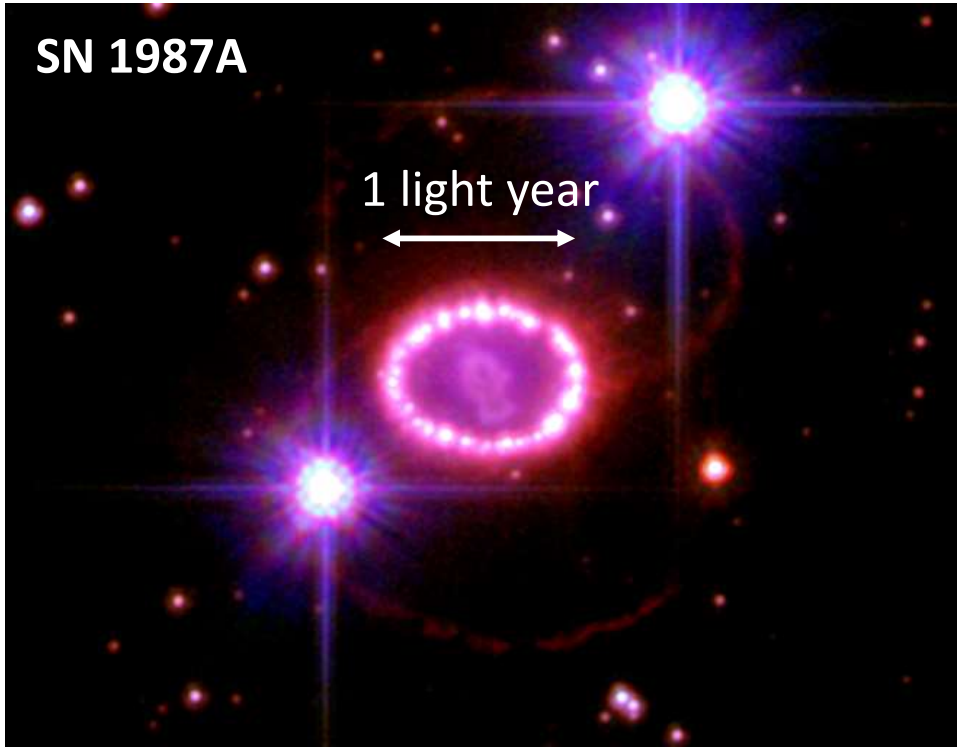
- Si burning produces ^{56}Fe – end of fusion process.
- Iron core undergoes gravitational collapse, leading to a supernova explosion.
- Atoms in the expanding star are bombarded with neutrons, leading to rapid (r-process) neutron capture and build-up of elements heavier than iron.



Crab Nebula: SN witnessed in 1054

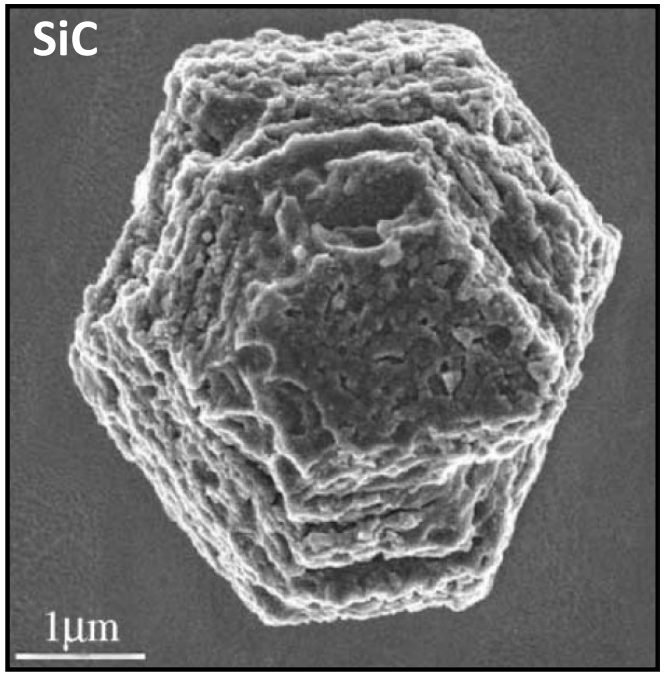
NASA/HST

SUPERNOVAE RECORDED IN PRESOLAR GRAINS IN METEORITES

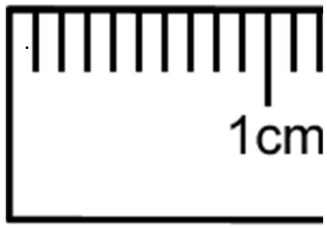


HST, PRC98-30

- As matter is ejected from a supernova, solid mineral grains condense (grow) from the gas.
- These grains are sprinkled into interstellar space.
- Some become trapped in meteorites.

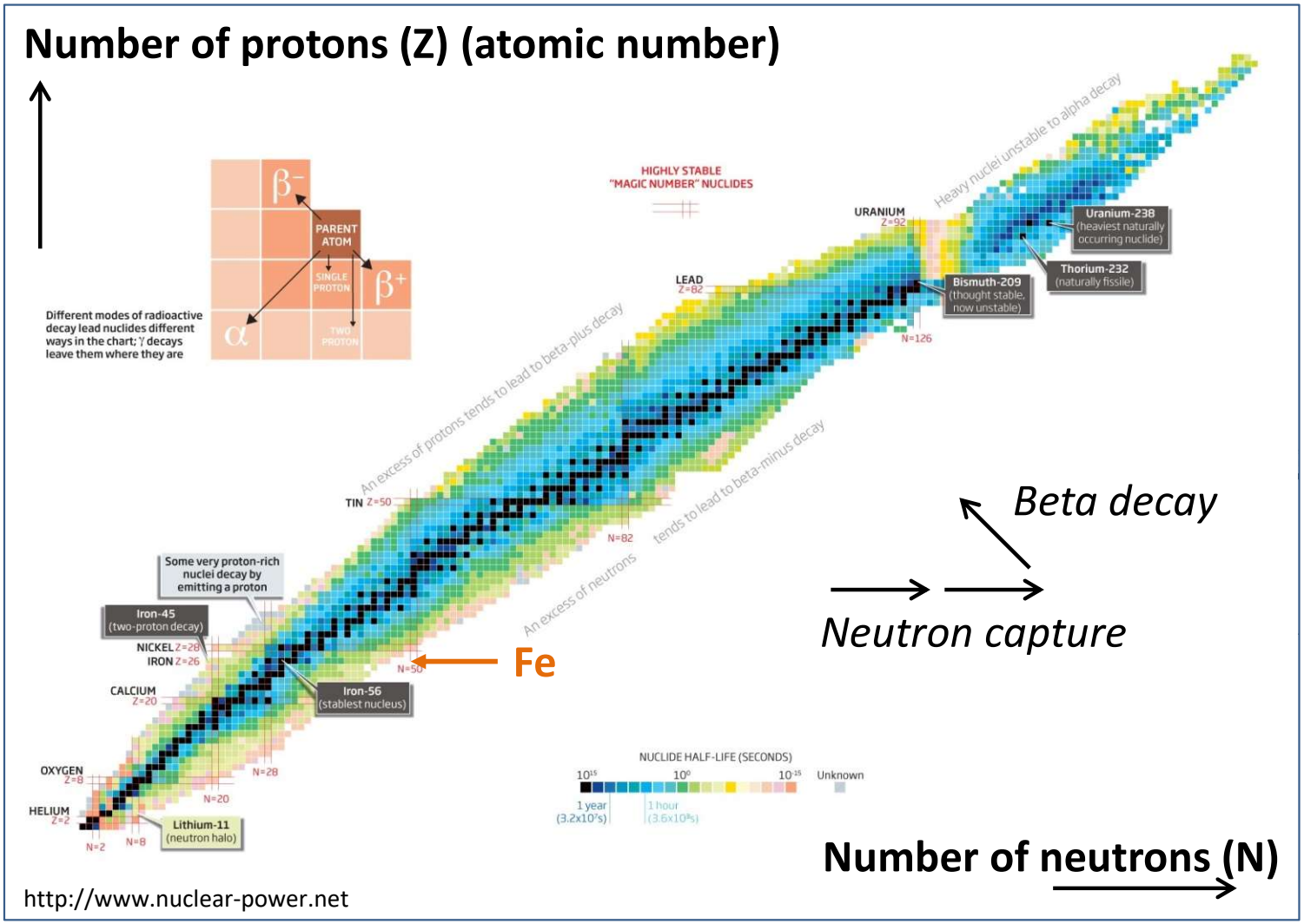


E. Zinner



A presolar grain of silicon carbide (SiC) extracted from a meteorite. The grain is a few micrometers (μm) across – much smaller than the dot on this cm scale.

NEUTRON CAPTURE: CHART OF THE NUCLIDES

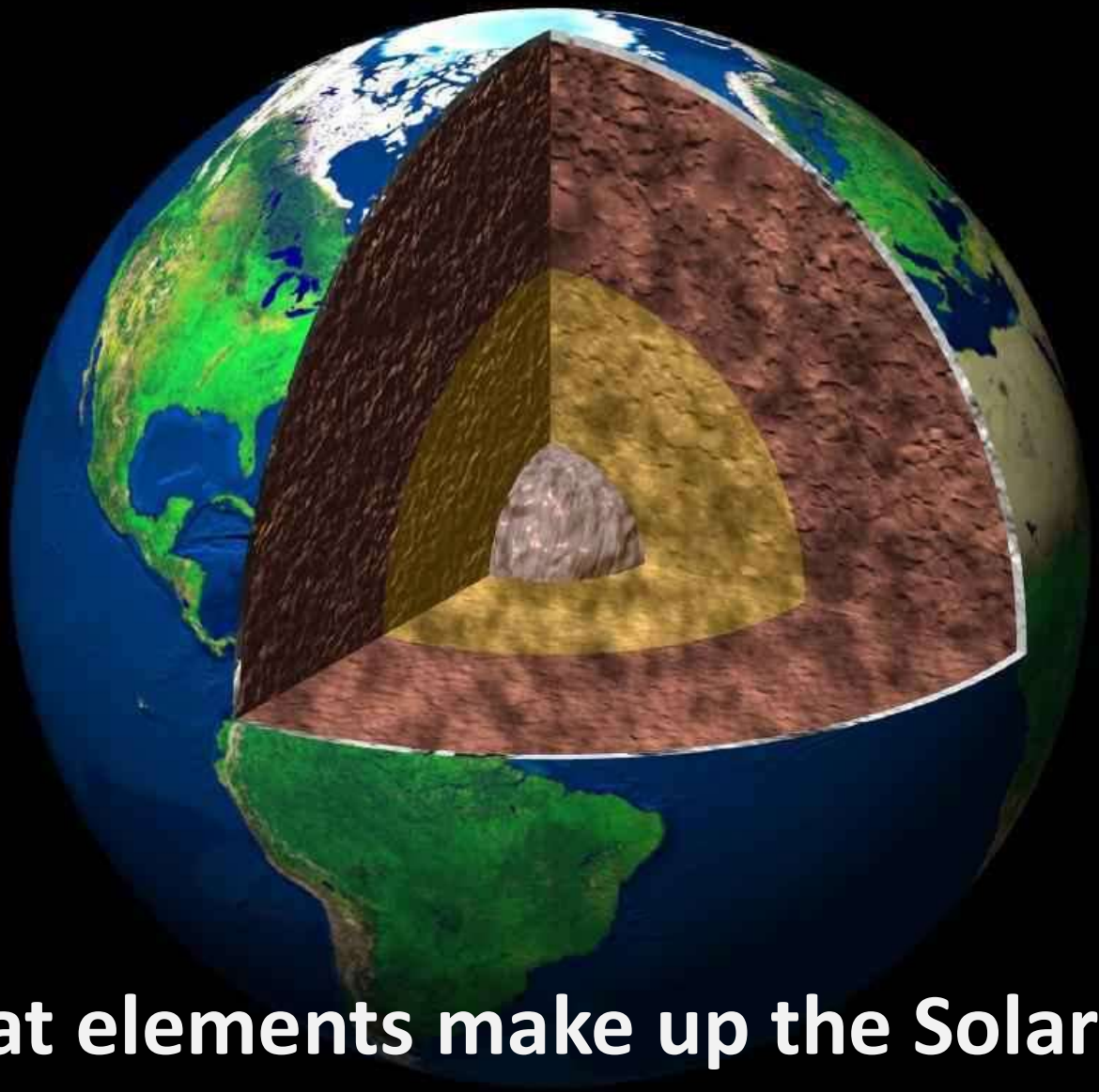


- Chart shows known isotopes. Black squares are stable isotopes.
- Neutron capture: adding neutrons to an element increases the mass of the element until it is no longer a stable isotope.

- Unstable nuclides, created from neutron capture, decay via beta-decay which moves the isotope up and left on the chart, forming a new element. Beta decay: $n \rightarrow p + e$
- s-process (slow-process) of neutron capture: build-up of nuclei heavier than ^{56}Fe in second-generation stars

Chemistry of the Earth: Lecture 1

What is the Earth made of?



Part 2: What elements make up the Solar System?

FORMATION OF THE SUN AND SOLAR SYSTEM (4.6 billion years ago)

1. Collapse of a dense region of a molecular cloud



HST: Eagle Nebula

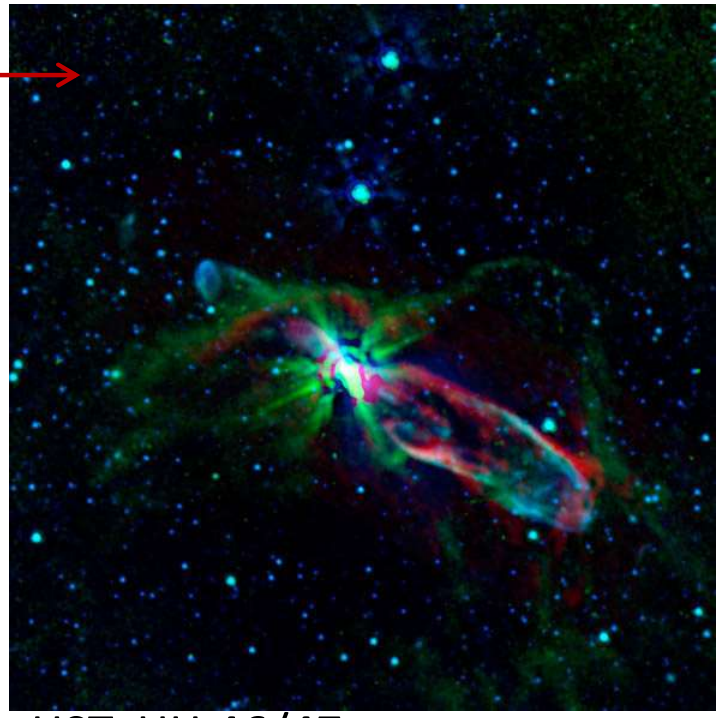
FORMATION OF THE SUN AND SOLAR SYSTEM

1. Collapse of a dense region of a molecular cloud



HST: Eagle Nebula

2. Formation of a new protostar with a disk and bipolar jet outflows



HST: HH 46/47

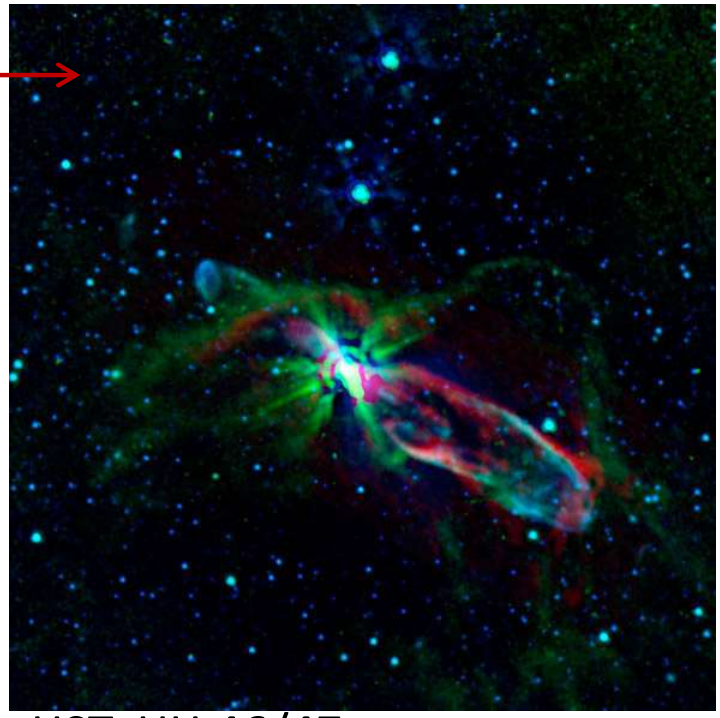
FORMATION OF THE SUN AND SOLAR SYSTEM

1. Collapse of a dense region of a molecular cloud



HST: Eagle Nebula

2. Formation of a new protostar with a disk and bipolar jet outflows



HST: HH 46/47

3. Protostar starts H burning reactions and becomes a star (the Sun). Rock and ice material condenses in the disk and accretes to form planets, satellites, asteroids, comets, etc.



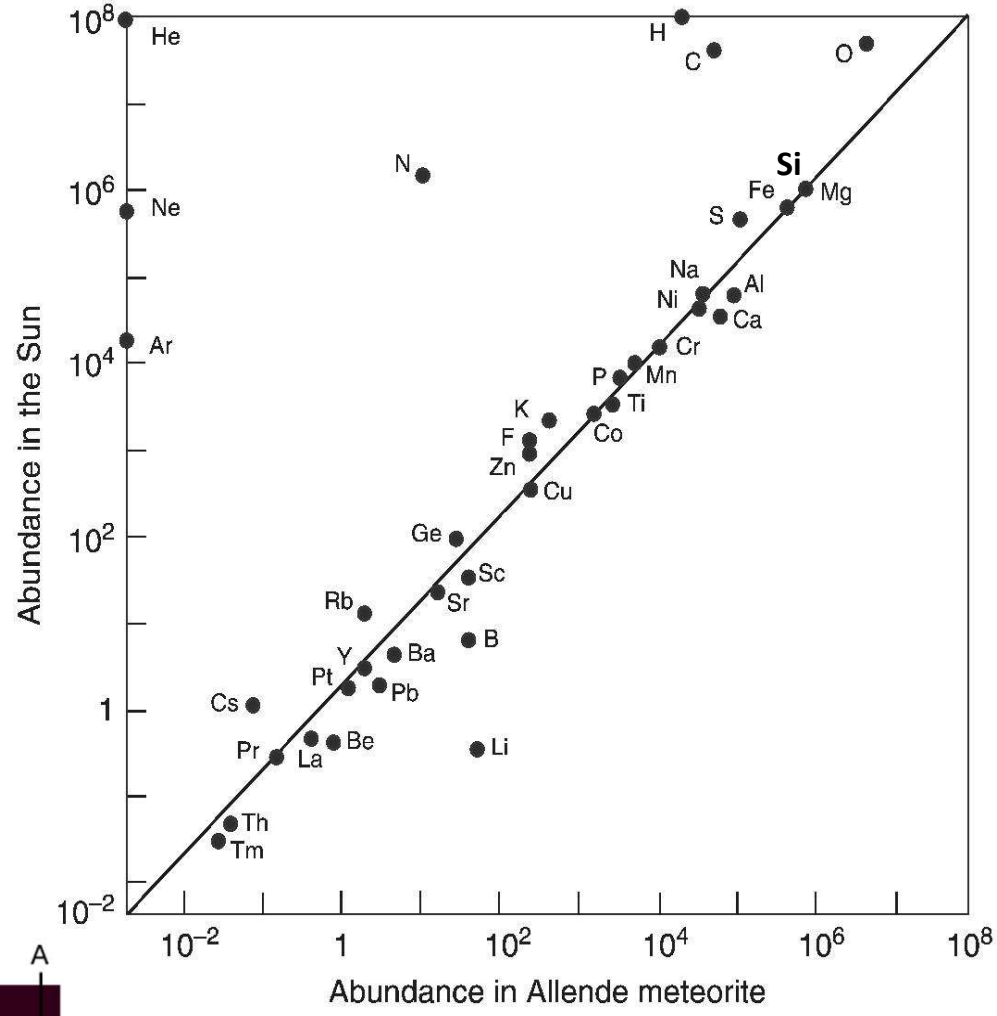
NASA

CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

Determined from chondrite meteorites and the composition of the Sun – these two are similar, except for elements that are usually gases.

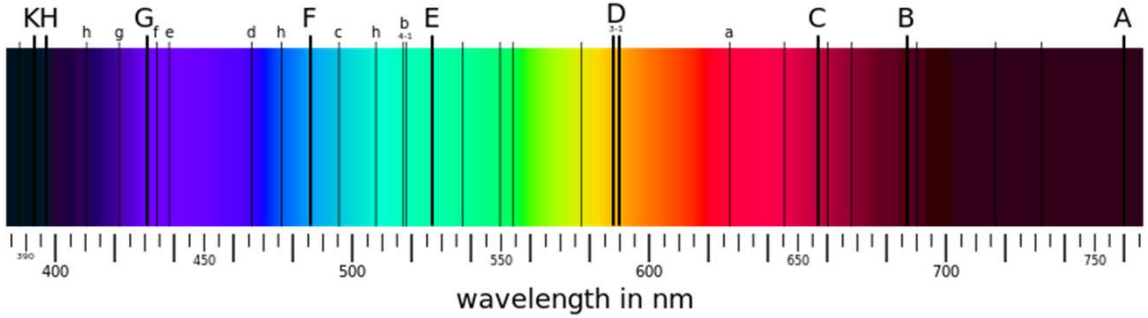


http://www.fallingrocks.com/Collections/Murchison.htm



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Sun – spectroscopy. Dark lines show absorption by different elements

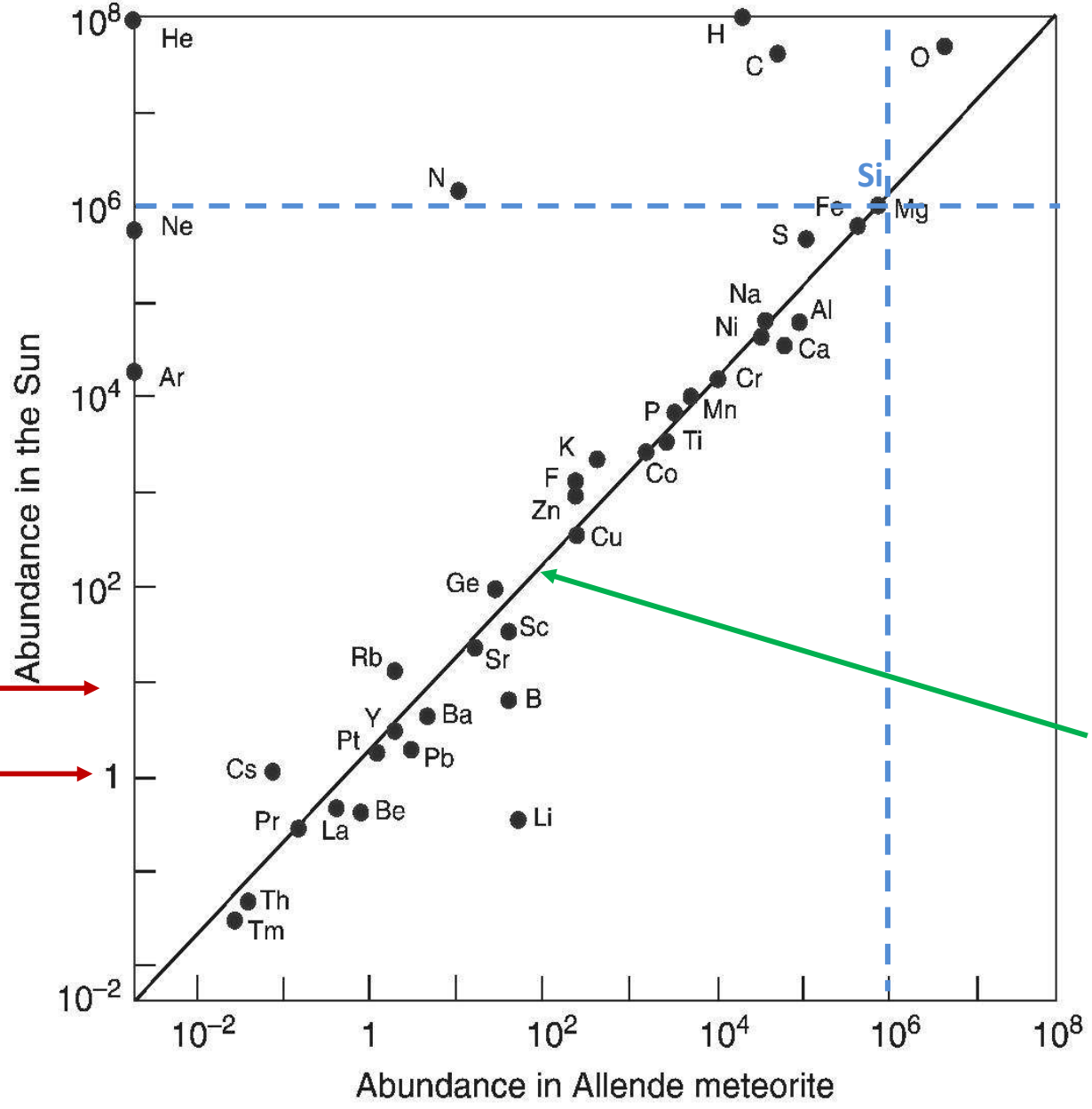


https://en.wikipedia.org/wiki/File:Fraunhofer_lines.svg

CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

Determined from chondrite meteorites and the composition of the Sun – these two are similar, except for elements that are usually gases.

Note:
Log scale



$10^1 = 10$

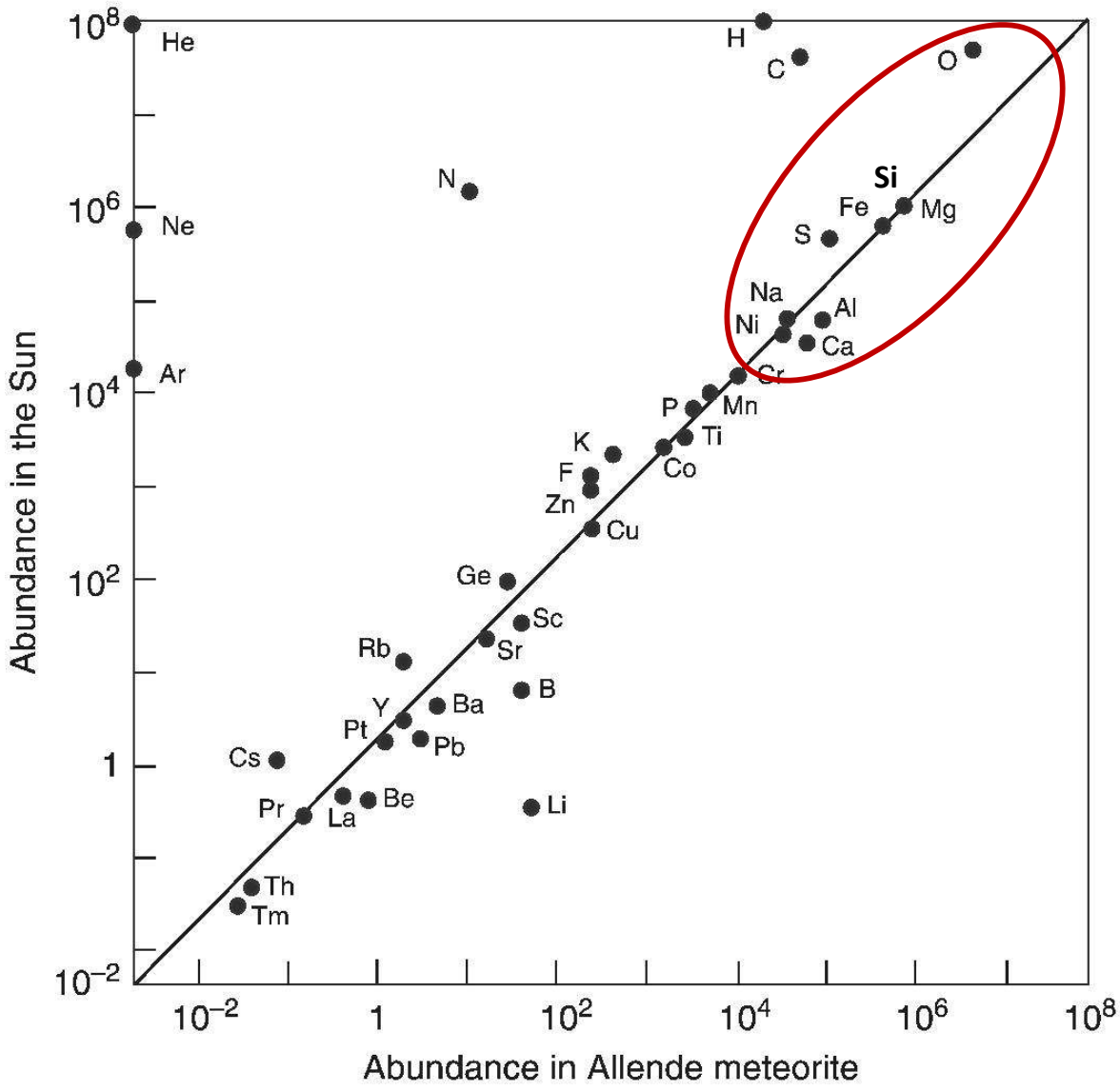
$10^0 = 1$

Abundances are plotted relative to 1 million silicon atoms ($Si = 10^6$)

Line shows 1:1 (equal) relationship

BULK COMPOSITION OF SUN / CHONDRITE

Most abundant rock-forming elements that make up the Earth



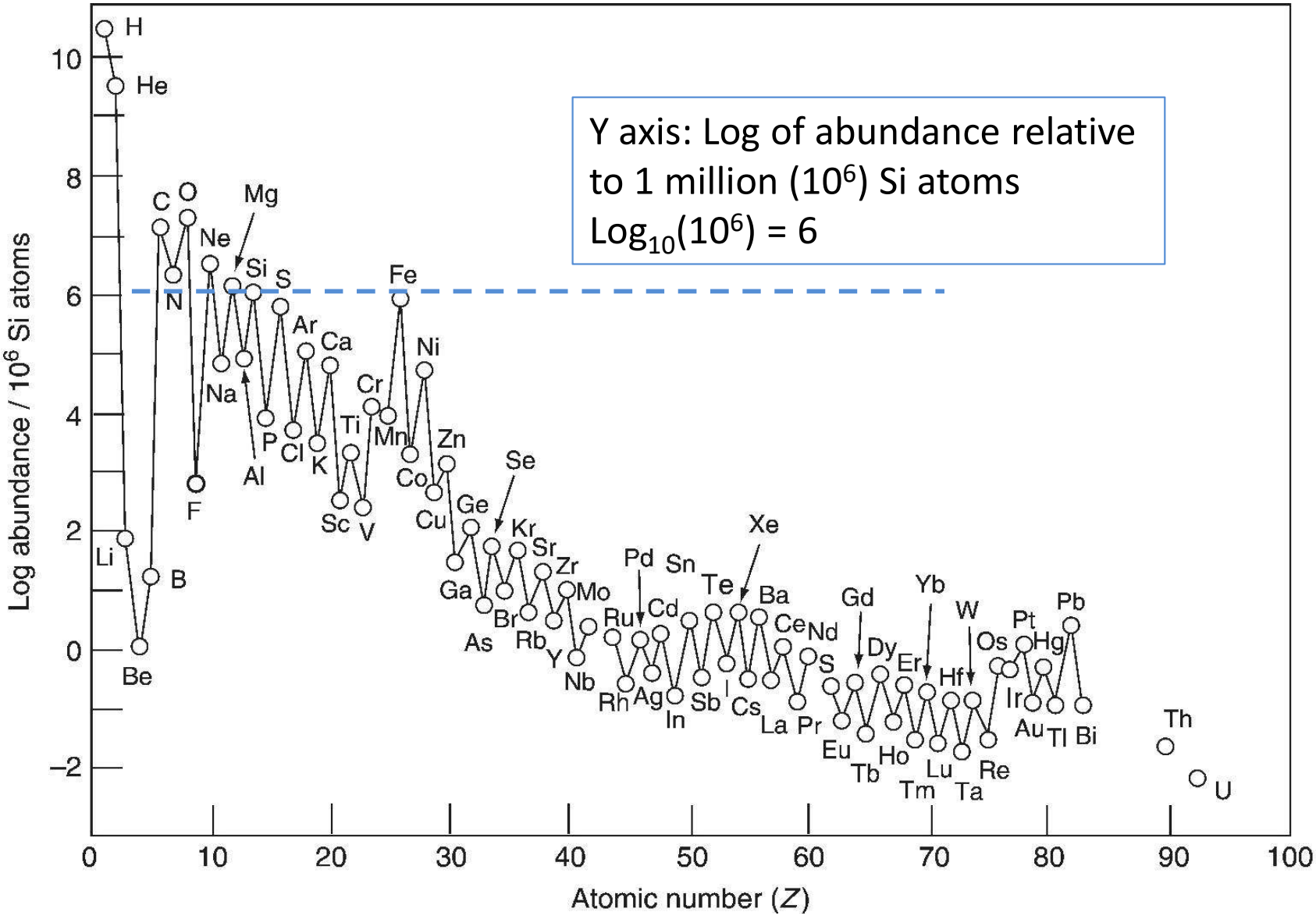
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O Oxygen
Mg Magnesium
Si Silicon
Fe Iron
S Sulphur

Al Aluminium
Ca Calcium
Na Sodium
Ni Nickel

CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

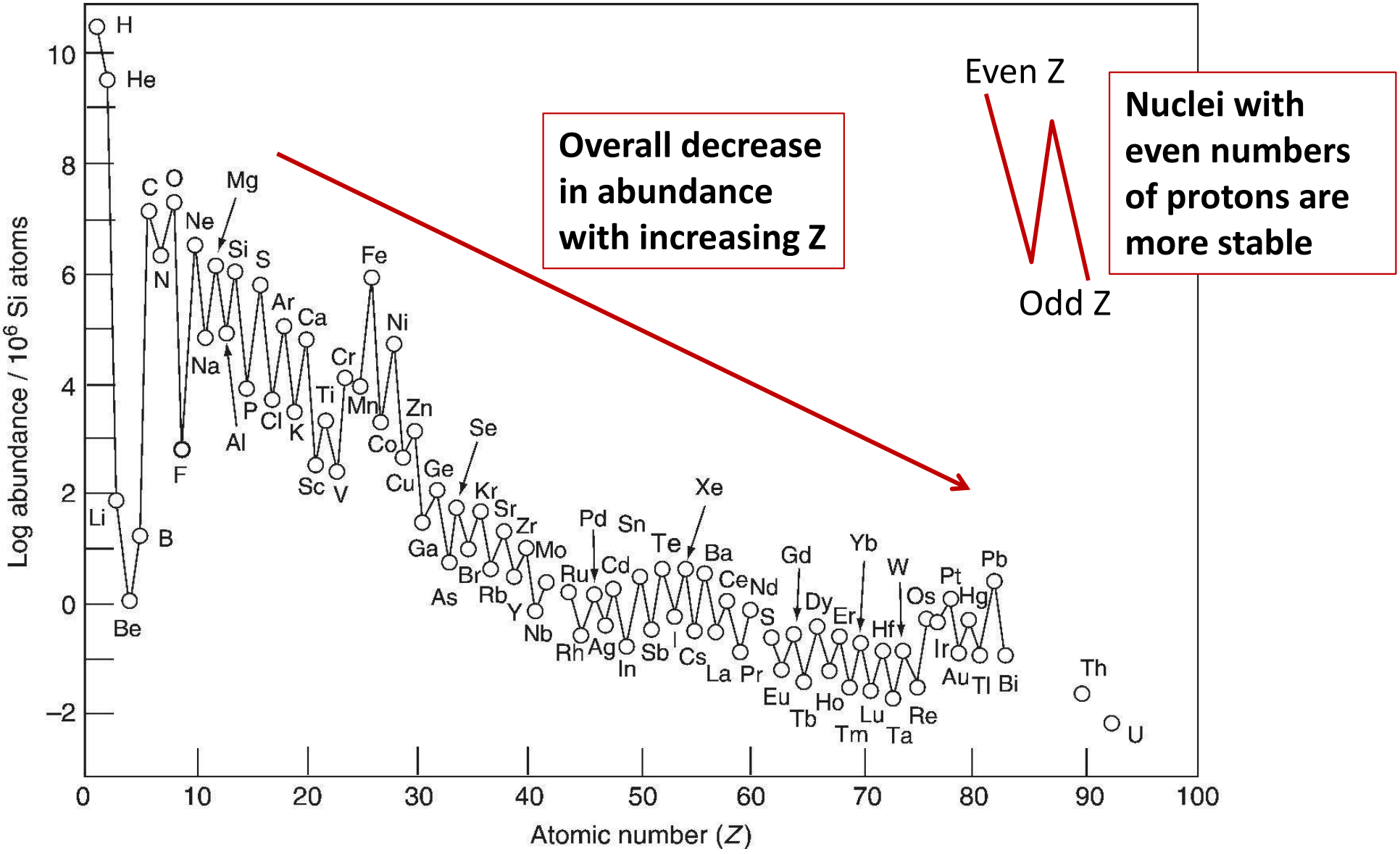
Composition makes sense when we know a little about nucleosynthesis



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CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

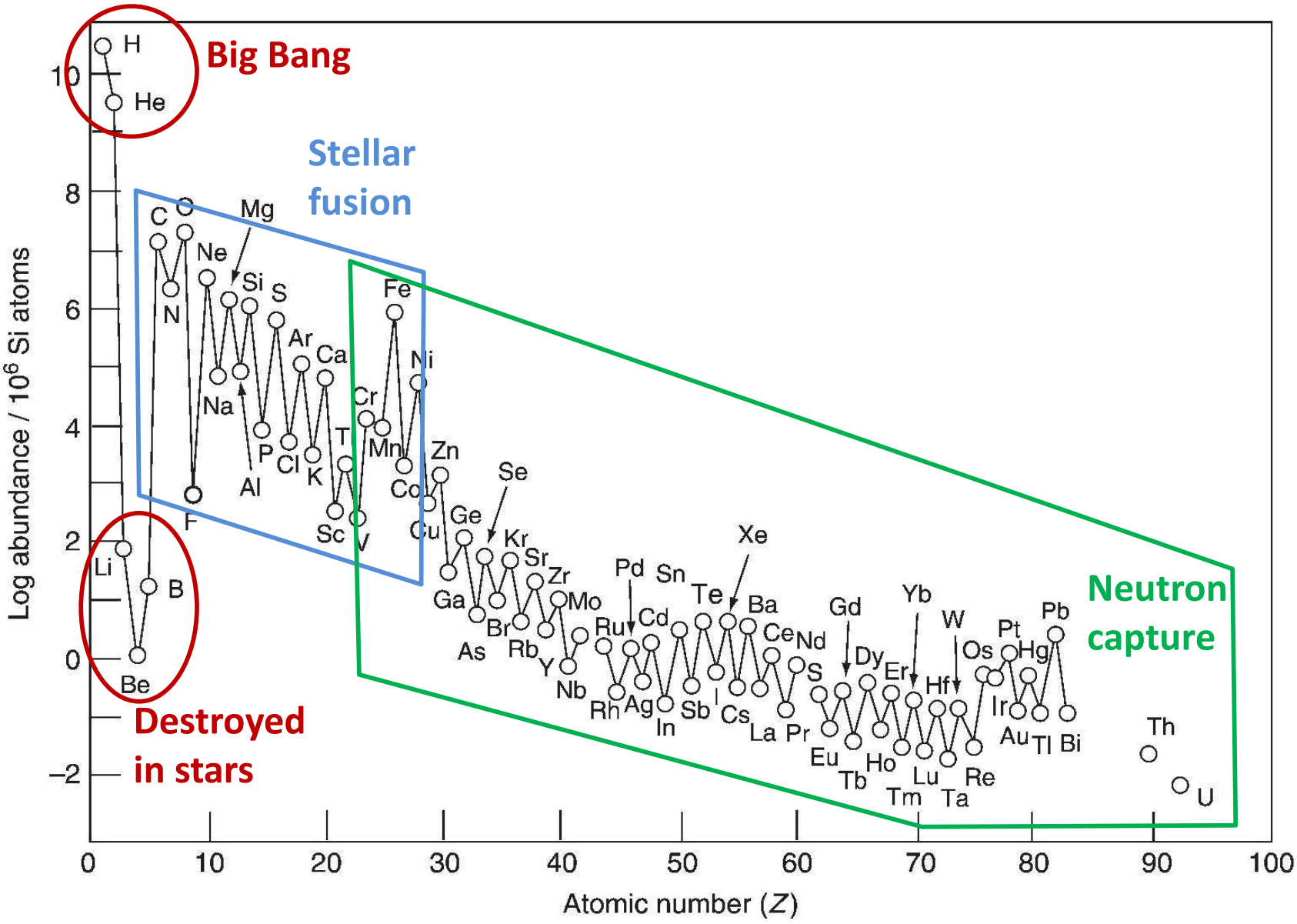
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CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

Composition makes sense when we know a little about nucleosynthesis

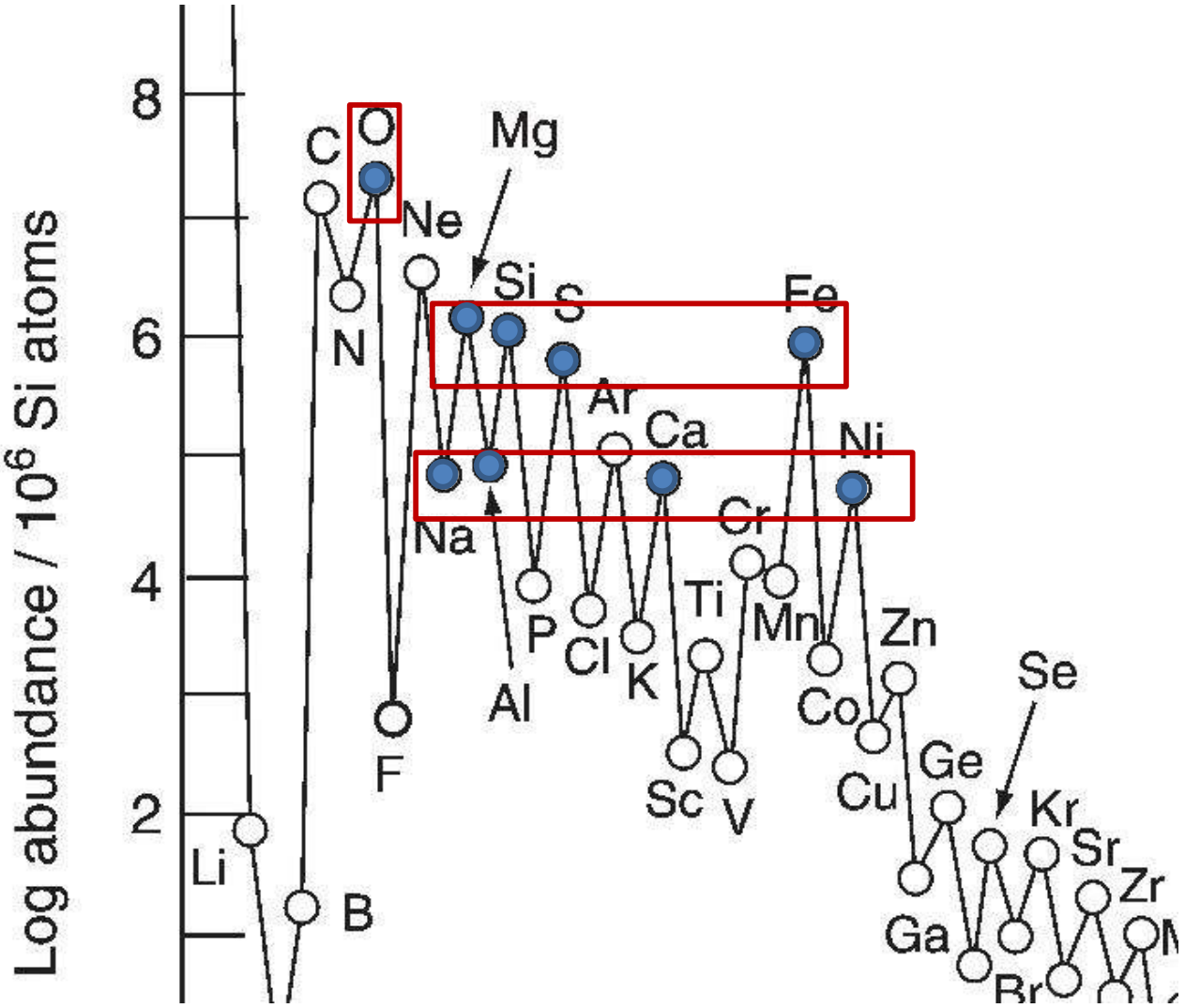


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MOST ABUNDANT ROCK-FORMING ELEMENTS

O Oxygen
Mg Magnesium
Si Silicon
Fe Iron
S Sulphur

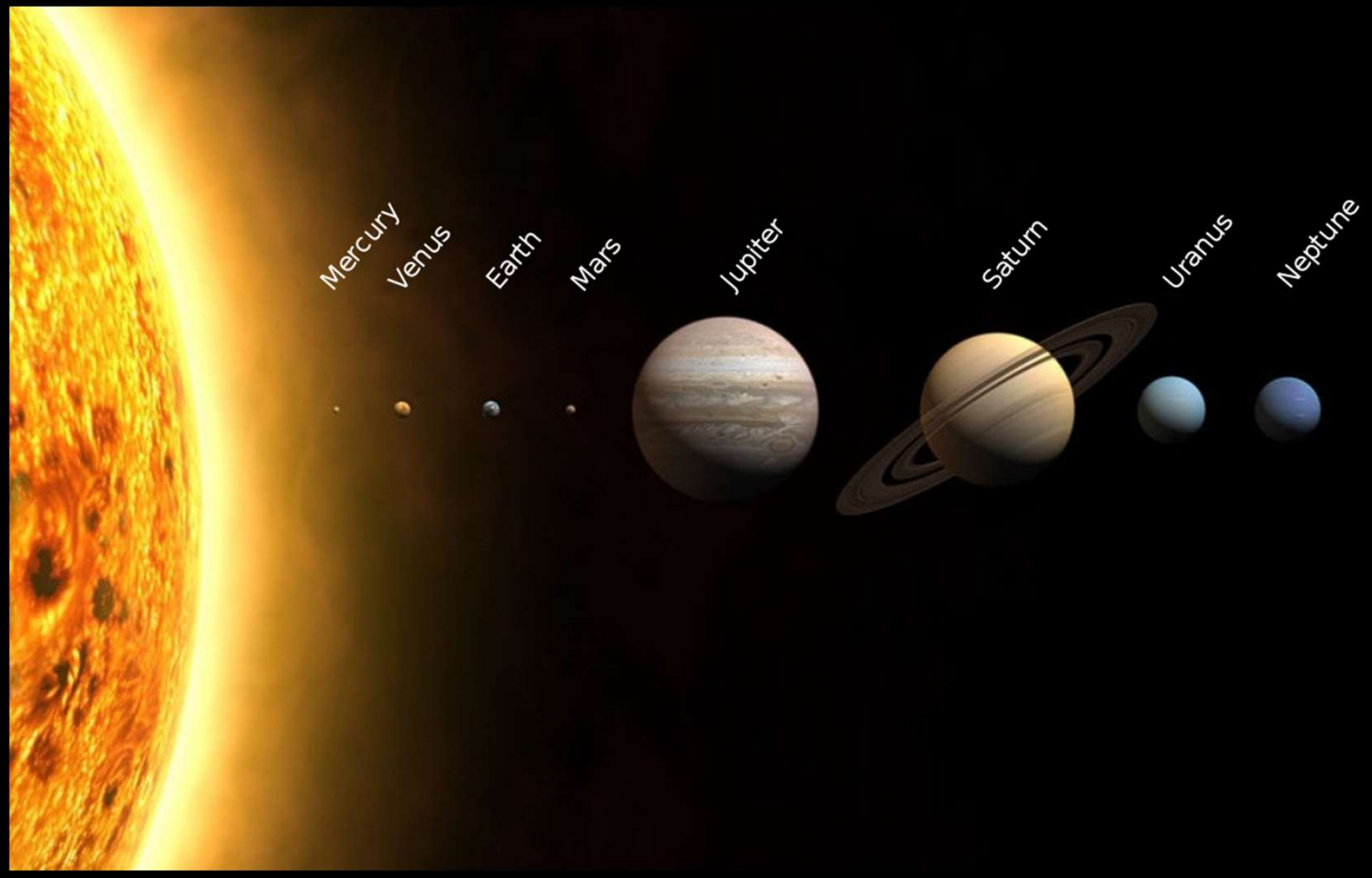
Al Aluminium
Ca Calcium
Na Sodium
Ni Nickel



(Ne and Ar are noble gases)

We have considered ……

- Where do elements come from?
- What is the composition of the Sun / Solar System?

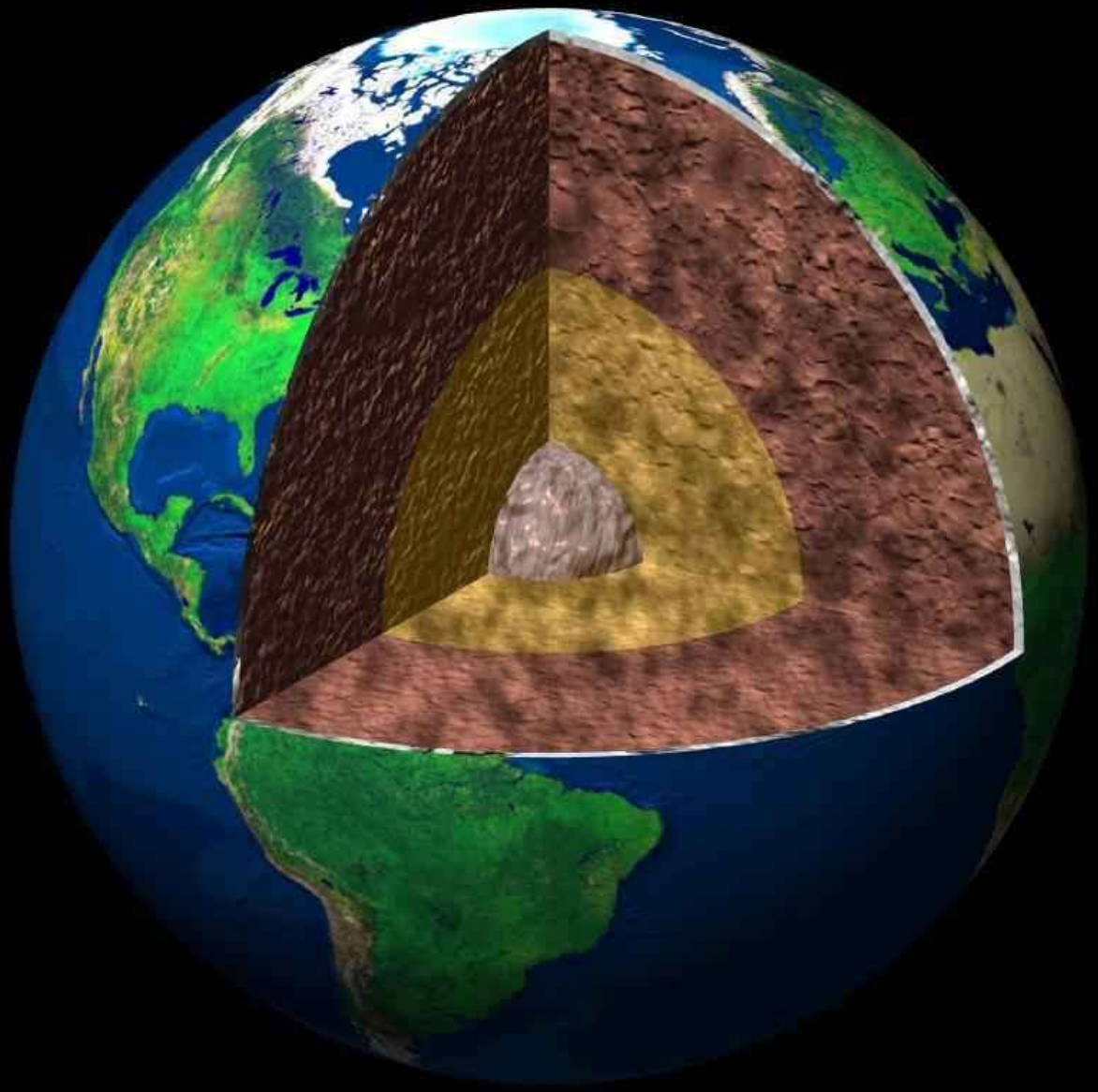


In the practical we will investigate ……

- What is the composition of the Earth?

Chemistry of the Earth: Practical 1

What is the Earth made of?



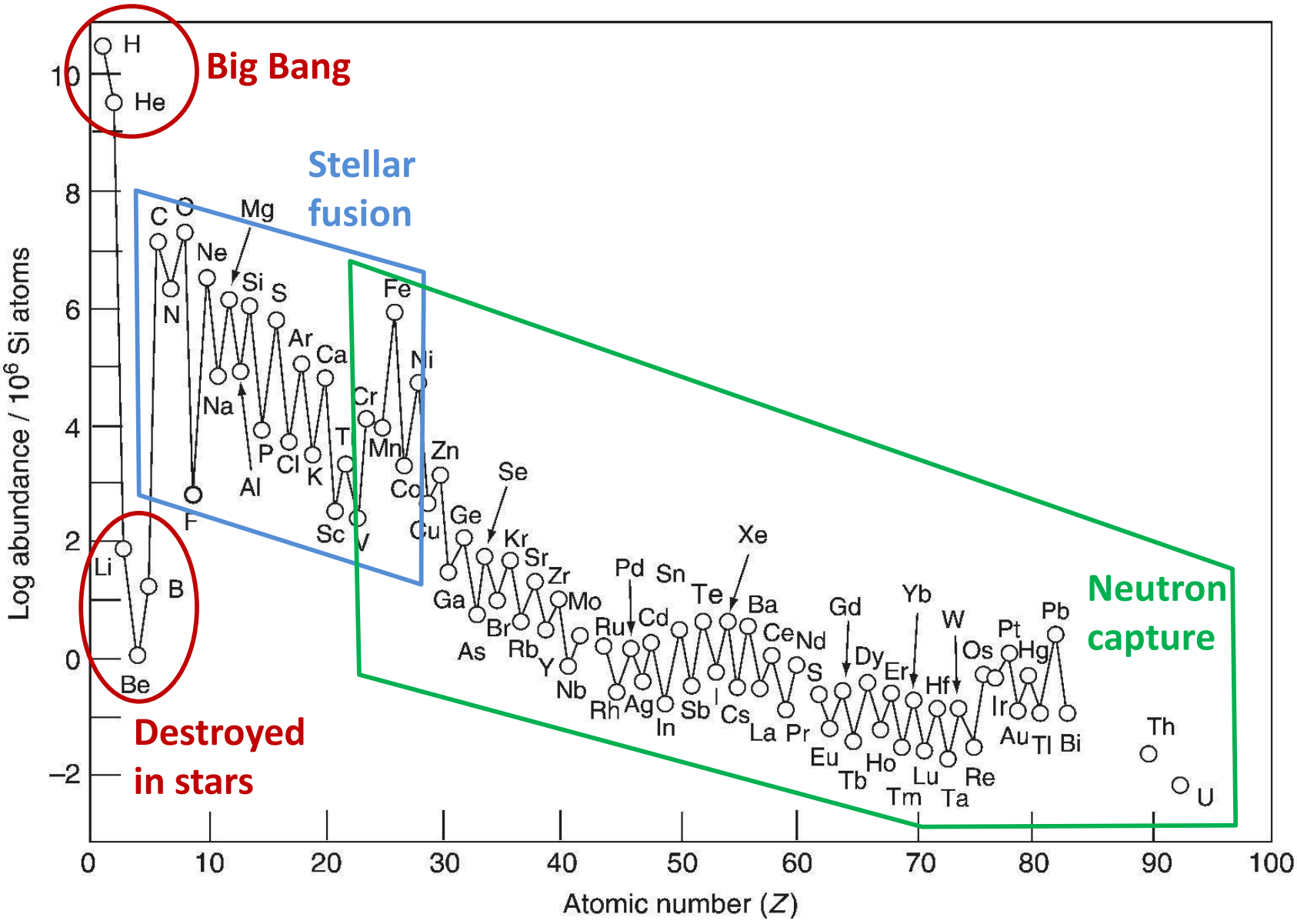
Practical 1:

What is the Earth made of?

- Review which elements are the most abundant in the Solar System, and why
- Investigate how the most abundant elements are distributed between the Earth's Core and Mantle
- Calculate the chemistry of the whole Earth based on Core and Mantle compositions
- Compare the composition of the whole Earth with the composition of the Solar System

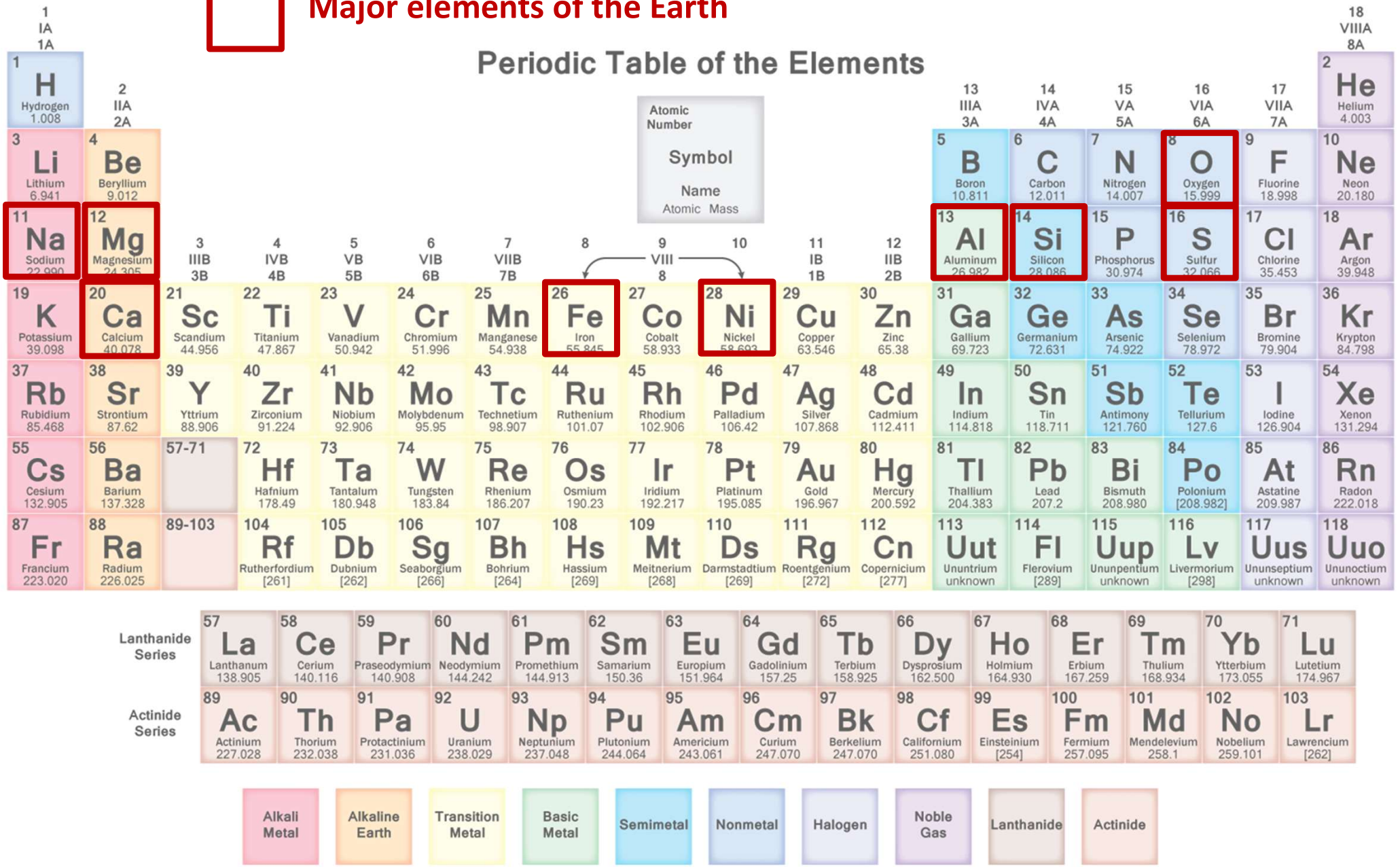
CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

Composition makes sense when we know a little about nucleosynthesis



Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
© 2012 Kula C. Misra. Published 2012 by Blackwell Publishing Ltd.

Major elements of the Earth



http://sciencenotes.org/printable-periodic-table/

CHEMICAL COMPOSITION OF THE EARTH

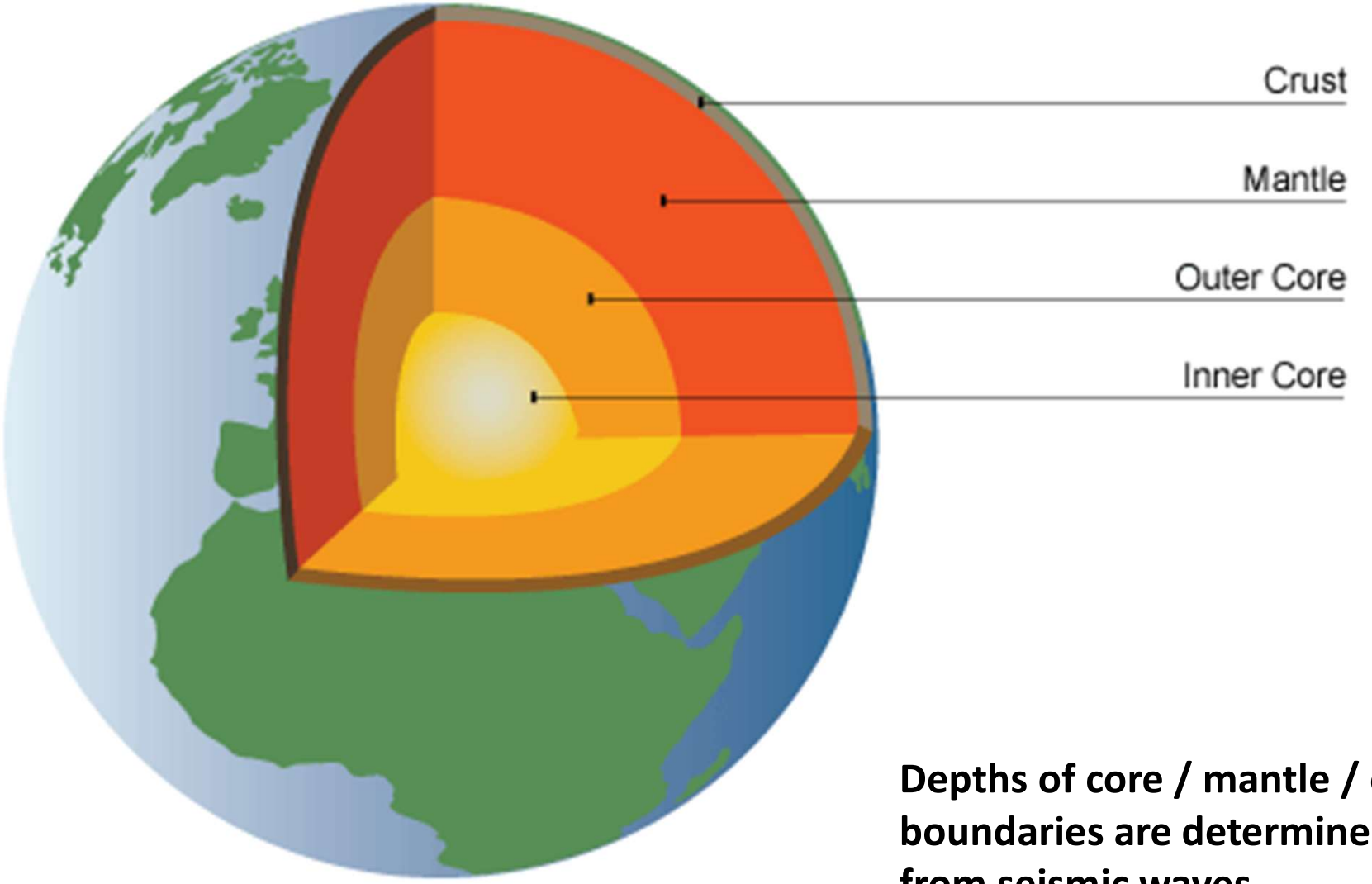
- Now we know the most abundant elements, but how much of each element is there?
- Is the Earth the same composition as chondrites and the Sun?



http://visibleearth.nasa.gov

BULK COMPOSITION OF EARTH

Try to piece it together by looking at different layers



Depths of core / mantle / crust boundaries are determined from seismic waves.

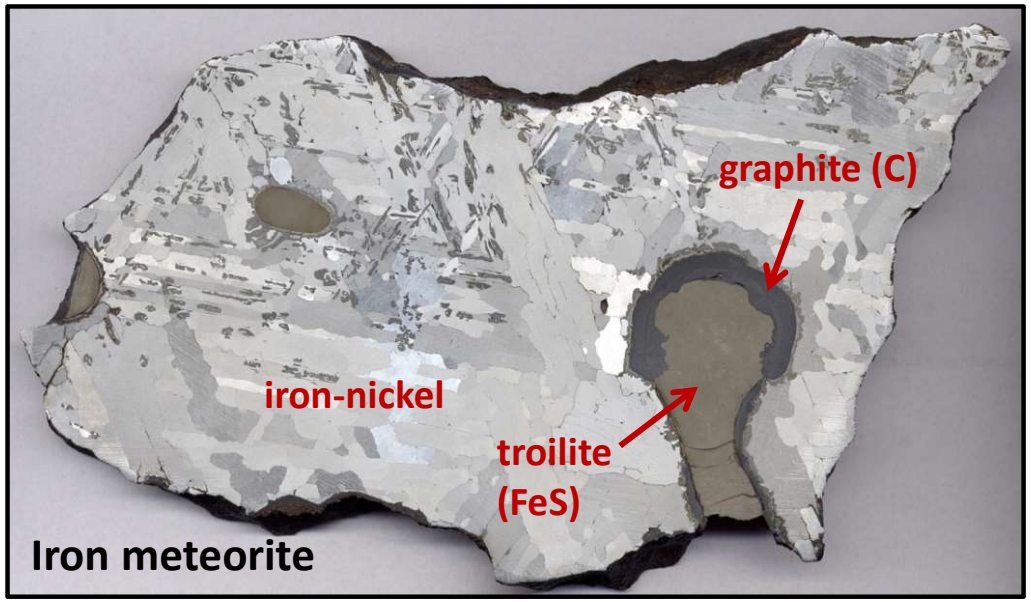
COMPOSITION OF THE CORE

• Approx half of Earth's radius.

• Must be dense to account for high density of planet compared to mantle / crust: core is 16% by volume and 31% by mass.

• High density inferred from seismic data and moment of inertia shows density concentrated at centre.

• Iron meteorites – Fe, Ni



Andrzej S. Pilski

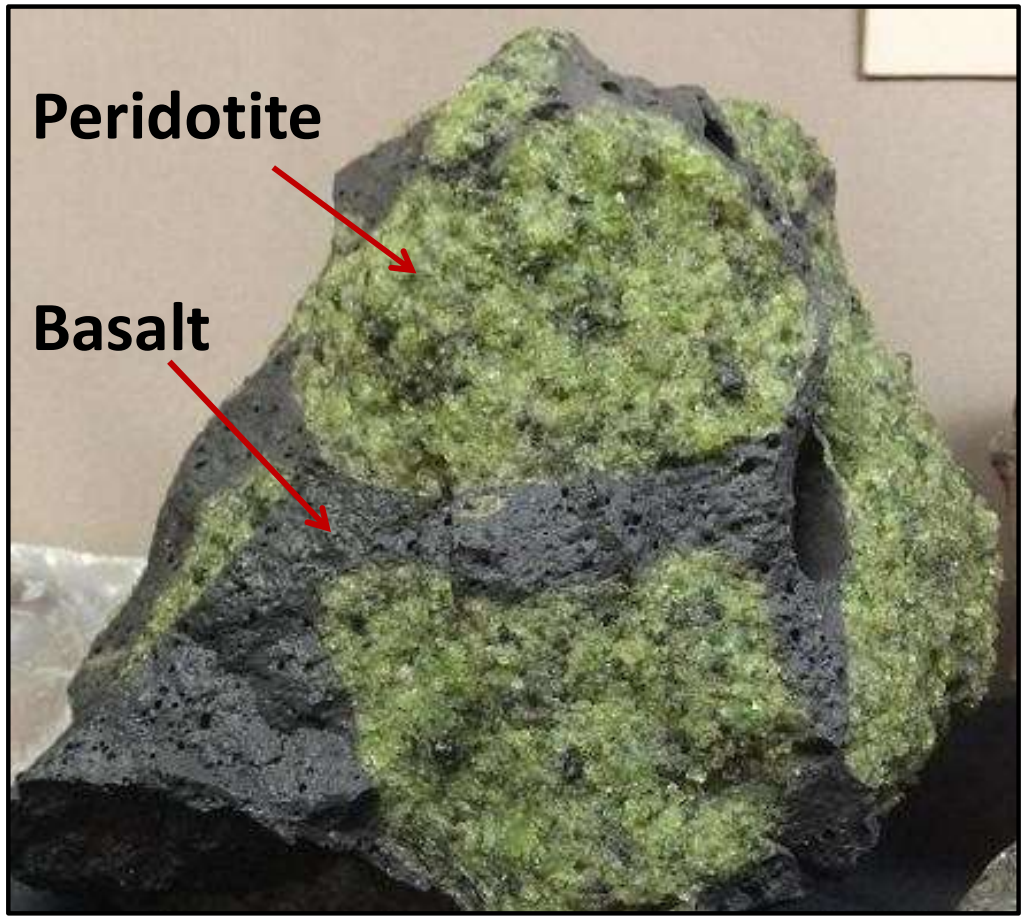
• But the core is not quite as dense as only Fe and Ni – S, O, Si?

Composition of Core (wt%)	Fe	Ni	S	O	Si
Misra - Kargel & Lewis (1993)	85	5	2	5	
Misra - Allegre et al. (1995)	85	5	2	4	7
McDonough (2013)	85	5	2	0	6
Rubie et al. (2011)	84	5	2	0.5	8

Inner core – solid
Outer core – liquid
Same composition?

COMPOSITION OF THE MANTLE

- Composition and structure determined from behaviour of seismic waves – Peridotite
- Must be capable of producing basalt upon partial melting – Peridotite
- Mantle xenoliths – Peridotite



http://peridot1.blogspot.co.uk/

Peridotite:

- Olivine and pyroxene – silicate (Si,O) minerals containing Fe, Mg and Ca.
- Fits with most abundant rock-forming elements.

Approximate mantle composition – major elements.
Given as wt% element.

O	44
Mg	23
Si	21
Fe	6
Ca	3
Al	2

McDonough 2013;
See also Albarède App. A

COMPOSITION OF THE CRUST

- Only about 0.5% of Earth's mass
- Accessible to geologists, so we know more about what it is made of
- Oceanic crust – thin (~6 km) – young – basaltic
- Continental crust – thick (~35 km) – old – very heterogeneous, average "andesitic" (more Si, Al, Na, K than basalt)



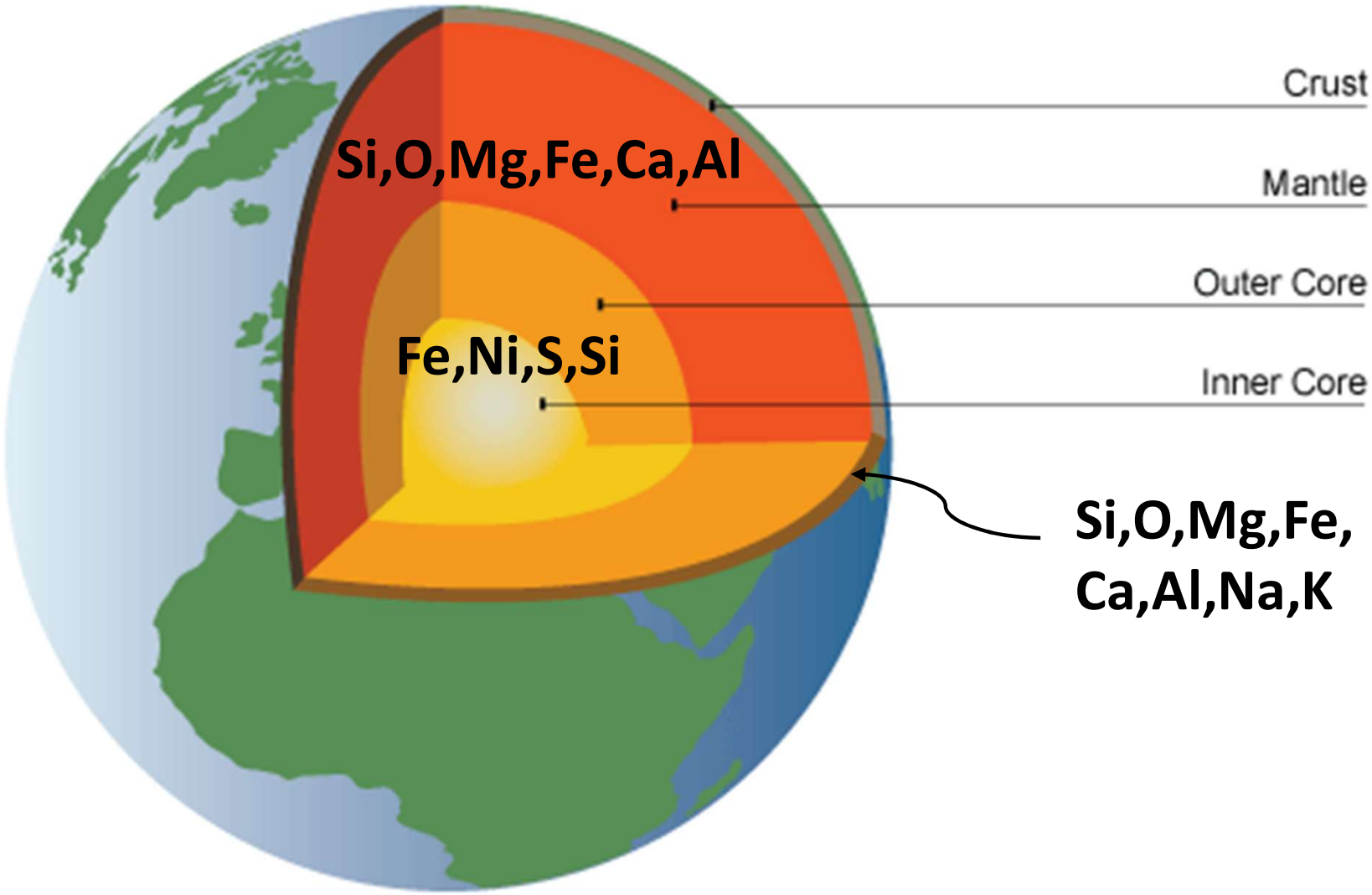
www.bgs.ac.uk

Approximate continental crust composition – major elements. Given as wt% element.

O	45
Si	28
Al	8
Ca	5
Fe	5
Mg	3
Na	2
K	2

see Albarède Appendix A

BULK COMPOSITION OF EARTH: Major elements in the different layers



Now do the practical exercise …….

Maths Concepts for today

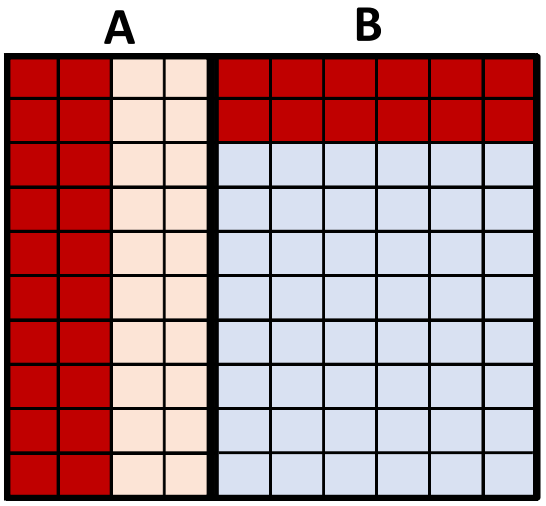
Volume of a sphere = $\frac{4}{3}\pi r^3$

Density = $\frac{\text{Mass}}{\text{Volume}}$

Mass Balance:

In the grid of 100 squares, what % of squares are red?

Box A:
50% is red



Box B: 20% is red

Red = 50% of A and 20% of B
Red = (50/100 * A) + (20/100 * B)
Red = (0.5 * A) + (0.2 * B)

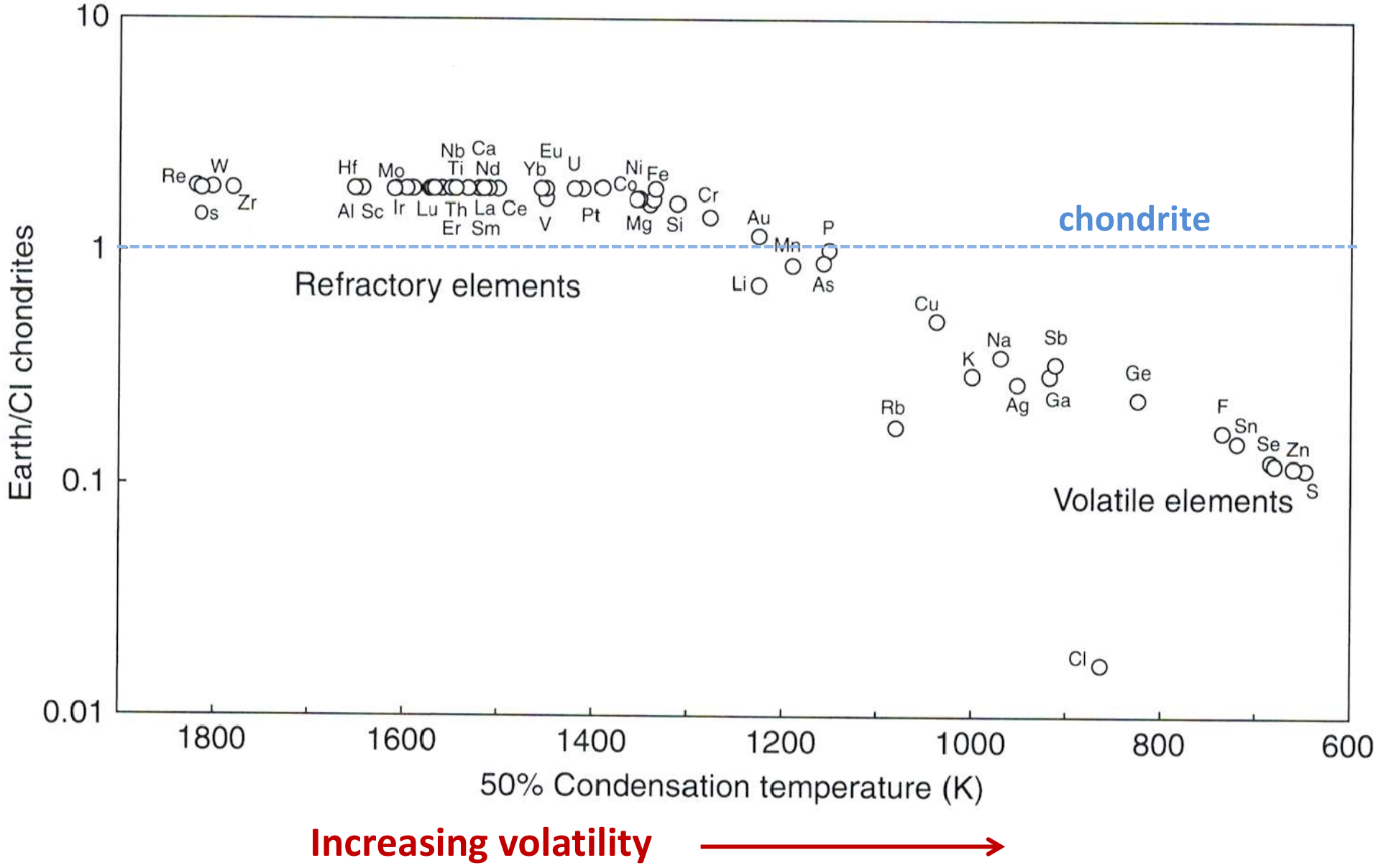
Box A is 40% of the grid; Box B is 60% of the grid

Red = (0.5 * 40%) + (0.2 * 60%)
Red = 20% + 12%
Red = 32% of the grid
(Because there are 100 squares, red = 32 squares)

Mass balance equation:
Total mass % = (fraction in A * % of A) + (fraction in B * % of B)

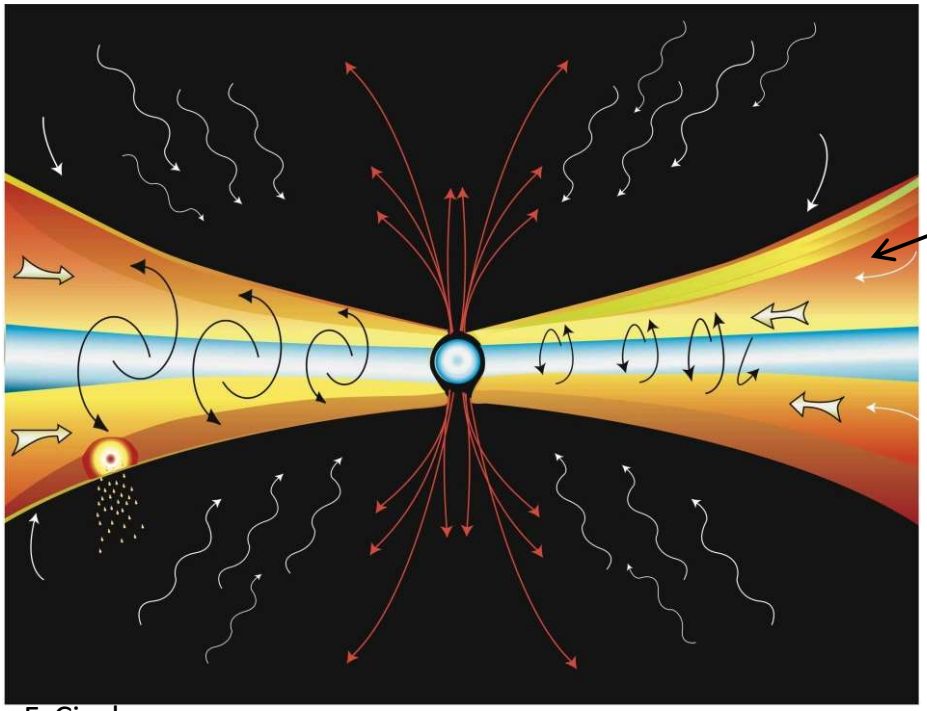
BULK COMPOSITION OF EARTH

• Overall, Earth is depleted in volatile elements compared with chondrites and the Sun.



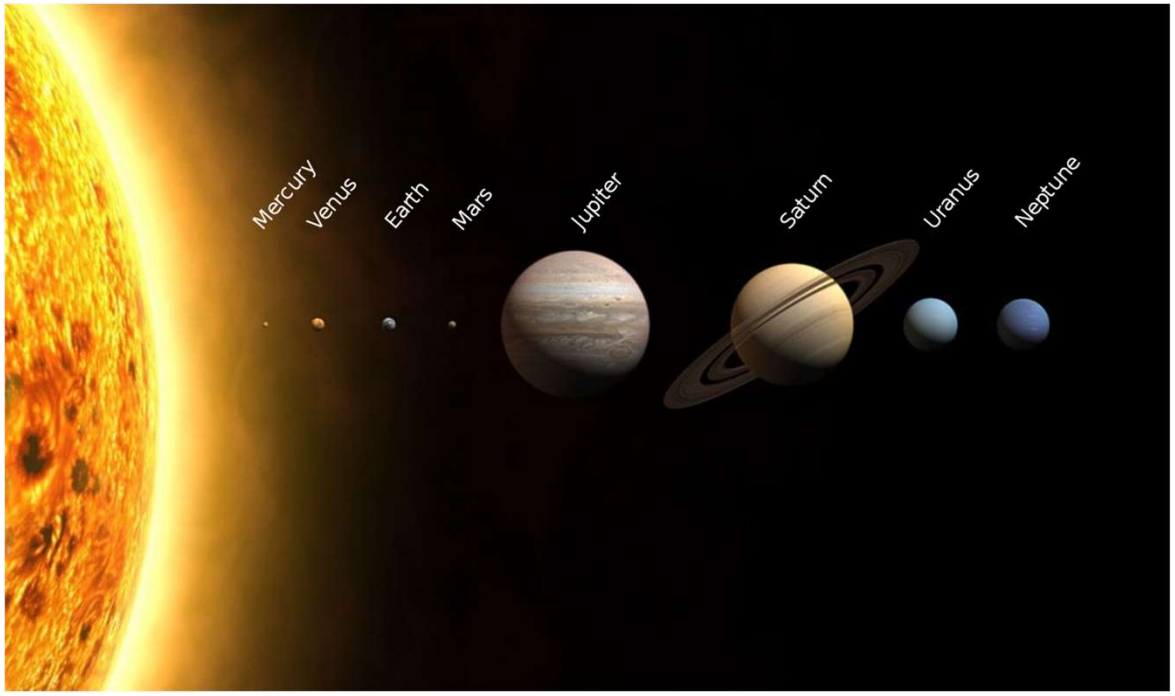
Albarede 2009 Fig. 12.11

Solar System formation



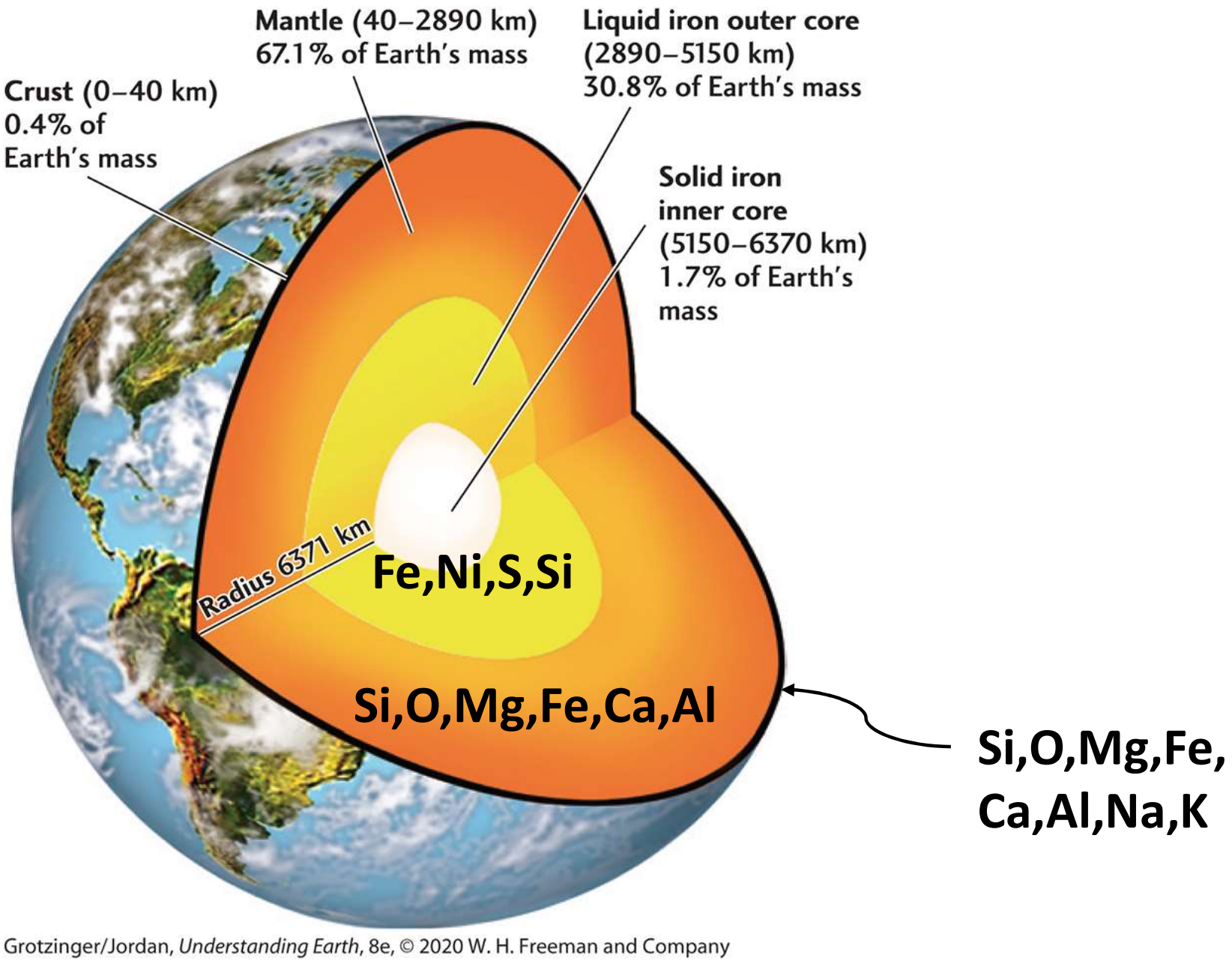
F. Ciesla

Protoplanetary disk:
Hotter closer to the Sun



For planets that formed closer to the Sun, volatile elements were still in the gas phase.

BULK COMPOSITION OF EARTH: Major elements in the different layers



Grotzinger/Jordan, *Understanding Earth*, 8e, © 2020 W. H. Freeman and Company